Screen Microphysics, Patches, and Observer Synchronization in OPH

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Abstract

This paper gives the fixed-cutoff implementation picture behind Observer-Patch Holography. Patches, overlaps, records, and observers are written as explicit finite registers with declared readout, repair, and synchronization rules. The result is a concrete fixed-cutoff screen architecture that can be simulated directly and used to analyze measurement, stable records, checkpoint and restoration, and observer synchronization inside one shared quantum system. The same architecture also gives the edge-sector heat-kernel law that feeds the gauge-coupling branch. On the quantitative side carried by the companion papers, the total, bulk, and edge entropy hierarchy has one exact self-similar balance point at the golden ratio, while an observer-supporting universe requires a small detuning away from that symmetry. This paper presents the finite reference surface on which the rest of the OPH program is built.

Contents

1 Scope

This paper sets out one concrete reference architecture for local finite screen degrees of freedom. It gives an explicit bridge from local registers to patches, overlaps, records, observers, and synchronization, together with fixed-cutoff theorem packages for measurement and checkpoint and restoration. The claim boundary is narrow: theorem statements, implementation targets, and imported companion-paper closure surfaces are separated throughout. The paper does not identify a unique microscopic UV completion, does not re-derive the companion-paper support-visible BW, geometric-modular, or local-Einstein closure surfaces, uses "observer" operationally, and does not present synchronization of overlap data as full refinement-limit closure.

2 Claim Taxonomy

The claims in this paper fall into five types.

(i) **Reused OPH structure.** The OPH screen-net picture, overlap consistency language, local MaxEnt and refinement-stability branch, recoverable generalized entropy package, and the regulator-scale edge-center completion language are reused as background structure.

(ii) **Architecture claims.** Once one finite gauge-register reference architecture is chosen, the patch algebra, overlap algebra, edge-sector observables, record layer, and observer-level interfaces can be defined explicitly at fixed cutoff.

(iii) **Fixed-cutoff implementation claims.** Small-lattice digital or tensor-network realizations can test whether the reference architecture exhibits gauge-invariant patch observables, stable records, low CMI across collars, and robust synchronization under local repair schedules.

(iv) **Imported companion-branch claims.** The support-visible BW, geometric-modular, and local-Einstein closure surfaces are supplied in the companion corpus; this paper only supplies the fixed-cutoff implementation interface that those surfaces use.

(v) **Scope-separated claims.** This paper does not identify the final OPH microphysics. The consensus paper carries finite repair and refinement-consensus theorem surfaces, and the companion papers carry the Standard Model and geometric-modular closure surfaces.

3 Why a Finite Gauge-Register Reference Architecture

Several microscopic families are logically possible: constrained finite gauge registers, quantum-link truncations of compact gauge systems, string-net or quantum-double models, generic tensor-network ansätze, or ad hoc observer automata attached to a screen. For a reference-model construction, the first working model should be selected by engineering criteria. Aesthetic breadth comes second.

The selection criteria are:

1. finite local Hilbert spaces at fixed cutoff;

2. exact local constraints;

3. a natural patch and overlap algebra;

4. explicit edge-sector or center observables on collars;
5. straightforward qudit-to-qubit compilation;
6. a path from local mismatch syndromes to repair or synchronization maps;
7. extensibility from toy groups such as $\mathbb{Z}_2$ or S_3 to richer quantum-link truncations.

A constrained finite gauge-register or quantum-link style screen model is therefore the default reference architecture for this draft. It is finite-dimensional, local, gauge-aware, and speaks the same language as the OPH regulator package. It also makes the collar center and edge-sector story concrete.

This choice is a working *reference architecture*. It leaves room for other microscopic families.

4 Reference Architecture: Local Screen Registers on a Cellulated Sphere

Take a finite cellulation $\Gamma = (V, E, F)$ of a horizon screen S^2. Each oriented link $e \in E$ carries a finite register $\mathcal{H}_e$. The simplest version is a finite-group register with basis $\{|g\rangle_e : g \in G\}$ for a finite group G, so

$$\mathcal{H}_e \cong \mathbb{C}^{d_G}, \qquad d_G = |G|. \tag{1}$$

A more flexible version replaces $|g\rangle$ by a truncated quantum-link basis labeled by a small irrep set and matrix indices. For phase 1, the finite-group presentation is easier to simulate, while the quantum-link presentation shows how the same pattern could extend to compact-group truncations.

To support records and observer logic, attach a small record register to selected vertices or coarse cells:

$$\mathcal{H}_v^{\mathrm{rec}} \cong (\mathbb{C}^2)^{\otimes m_v}. \tag{2}$$

The extended Hilbert space is then

$$\widetilde{\mathcal{H}}_\Gamma = \bigotimes_{e\in E} \mathcal{H}_e \otimes \bigotimes_{v\in V} \mathcal{H}_v^{\mathrm{rec}}. \tag{3}$$

At each vertex v, a local gauge action acts on the incident oriented links by left/right multiplication or by the corresponding truncated quantum-link action. Let P_v denote the Gauss projector onto the locally admissible subspace. The physical cutoff Hilbert space is

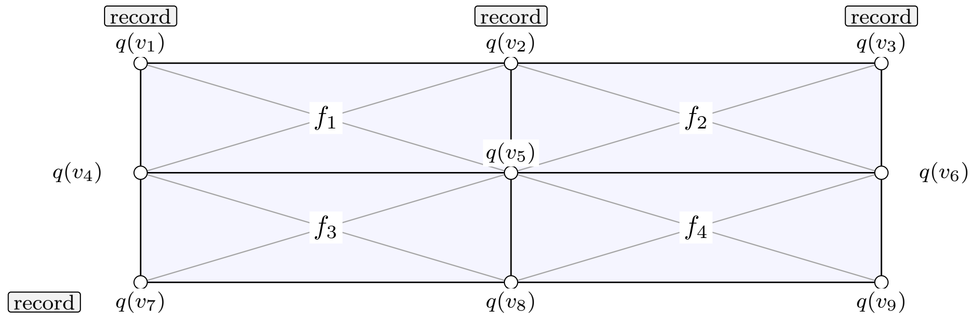
$$\mathcal{H}_{\mathrm{phys}} = \Big(\prod_{v\in V} P_v\Big)\widetilde{\mathcal{H}}_\Gamma. \tag{4}$$

A minimal reference Hamiltonian or Trotterized circuit family should contain four layers:

$$H_{\mathrm{ref}} = \lambda_G \sum_v (1 - P_v) + \lambda_B \sum_f (1 - B_f) + \lambda_R \sum_a (1 - R_a) + H_{\mathrm{sync}}, \tag{5}$$

where B_f are plaquette or face operators, R_a are record-stabilization or record-write terms, and H_{sync} is a local overlap-repair interaction to be chosen from a small menu later in the paper. The exact coefficients and update schedules are not fixed here. What matters is the architecture: gauge constraints, face dynamics, record dynamics, and repair dynamics all live on one finite screen.

The implementation question is direct. Can one finite local screen model realize the OPH patch, edge, record, and synchronization interfaces well enough to be testable? The physically meaningful uniqueness statement is quotient-level implementation hiding together with inert ancillary stabilization, not uniqueness of one screen-register presentation.



Links	finite gauge registers $\mathcal{H}_e$.
Vertices	Gauss checks P_v and optional record qubits.
Faces	plaquette or holonomy dynamics B_f.

Figure 1: Microscopic screen picture for the finite gauge-register reference architecture.

5 Qudit-to-Qubit Encoding Agenda

If the screen model is ever going to be built or simulated on conventional hardware, the local finite registers must be compiled into qubits. This paper does not pick a single global encoding forever; it sets the agenda and the tradeoffs.

5.1 Binary, unary, and factorized encodings

For a d-level link register, the default binary encoding uses

$$n_e = \lceil \log_2 d \rceil \tag{6}$$

qubits per link, with a penalty term suppressing unused binary strings when d is not a power of two. Binary encoding is qubit-efficient and natural for digital simulation.

Unary or one-hot encoding uses d qubits per link with one excitation allowed. It is less qubit-efficient but makes local projectors and permutation actions shallower. This is useful for very small groups and for verification runs.

For truncated quantum-link bases labeled by irreps and matrix indices, a factorized encoding is often cleaner:

$$|R, m, n\rangle \longmapsto |R\rangle_{\text{irrep}} \otimes |m\rangle_{\text{left}} \otimes |n\rangle_{\text{right}}. \tag{7}$$

This keeps the representation label explicit, which is attractive when collar sector labels are later measured.

5.2 Agenda for phase 1

The encoding work should proceed in the following order.

1. Start with $G = \mathbb{Z}_2$ or $\mathbb{Z}_N$ in binary encoding to validate the patch, overlap, and synchronization pipeline cheaply.

2. Move to S_3 or another small nonabelian finite group to validate nonabelian sector projectors, collar labels, and character-based observables.

3. Add a truncated quantum-link variant only after the finite-group version produces stable measurements and sensible scaling diagnostics.

4. Compare binary and unary encodings only at the level of compilation cost and numerical stability, not as a statement about final physics.

What must be compiled into qubit circuits is explicit:

1. local gauge transformations and Gauss projectors;
2. plaquette or loop operators;
3. record-write and record-read channels;
4. overlap projectors, edge-sector projectors, and mismatch syndromes;
5. local repair maps that preserve gauge admissibility.

6 From Local Registers to Patches and Overlaps

The bridge from local finite registers to OPH patch language has to be explicit. A patch is a connected geometric region together with the gauge-invariant observable algebra available there.

Let $P \subseteq \Gamma$ be a connected collection of faces, together with all vertices and links incident to those faces. Its extended cutoff Hilbert space is

$$\widetilde{\mathcal{H}}_P = \bigotimes_{e \subset P} \mathcal{H}_e \otimes \bigotimes_{v \subset P} \mathcal{H}_v^{\mathrm{rec}}, \tag{8}$$

and its extended local algebra is

$$\widetilde{\mathcal{A}}(P) = \mathcal{B}(\widetilde{\mathcal{H}}_P). \tag{9}$$

The physical patch algebra is the boundary-fixed subalgebra

$$\mathcal{A}(P) = \widetilde{\mathcal{A}}(P)^{G_{\partial P}}, \tag{10}$$

matching the regulator picture used throughout the framework.

If P and Q overlap, the shared region $B = P \cap Q$ is more than a geometric set. It is where the shared or comparable observables live. At fixed cutoff, the collar Hilbert space is expected to decompose as

$$\mathcal{H}_B \cong \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}), \tag{11}$$

with center projectors Π_α labeling edge sectors. That decomposition is not proved again here; it is reused OPH regulator language and serves as the observable interface for the reference model.

6.1 Patch and overlap observable dictionary

The working dictionary is the core of this paper.

1. **Link register basis / electric occupation.** The local computational basis on a link is the microscopic screen data. In a finite-group presentation this is the basis $|g\rangle_e$; in a quantum-link truncation it is the local representation basis.
2. **Gauss projector P_v.** The Gauss projector is the local admissibility check. It selects the neighboring link configurations that are physical around a vertex.

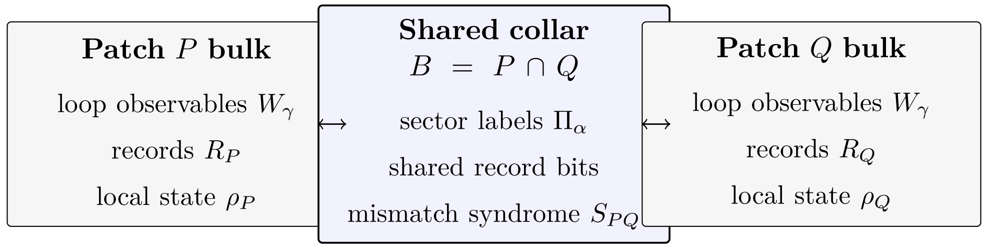


Figure 2: Patch/overlap interface. The collar carries only the selected shared observables used for comparison and synchronization, not the full patch quantum state.

3. **Plaquette operator B_f or loop operator W_γ.** The plaquette or loop operator probes curvature, flux, or sector excitation. Closed loops contained inside a patch belong to $\mathcal{A}(P)$.

4. **Boundary or collar sector projectors Π_α.** These are the overlap-accessible center observables. They are the natural candidates for shareable edge data and for robust cross-patch records.

5. **Record observables $Z^{\mathrm{rec}}_{v,a}$ and pointer subalgebras.** These are commuting or approximately commuting observables stabilized long enough to act as memories. A record is useful only if it can be read repeatedly with low back-action on the shared sector data.

6. **Patch state ρ_P.** The reduced state on $\mathcal{A}(P)$ is the patch description accessible to an internal pattern supported on P.

7. **Overlap mismatch syndrome S_{PQ}.** This is a function of shared observables, typically of the collar sector labels and the record comparisons on $B = P \cap Q$. It is the direct input to any repair or synchronization update.

What is compared on B is not the whole quantum state; it is a selected shared observable family on the overlap.

The observer interface is therefore local: persistent access to a patch algebra together with a stable shared observable family on overlaps is sufficient.

7 Records and the Observer Criterion

The word "observer" should be used operationally. Here, an observer is an internal pattern with patch access and records. In a finite screen model, that can be stated more sharply.

7.1 Records

A *record layer* is a commuting or approximately commuting subalgebra of robust pointer observables, supported partly inside a patch and partly on the boundary sector interface. A useful record variable must satisfy three practical conditions:

1. it is written by local interactions with nearby screen registers;
2. it survives for many local update cycles compared with the readout timescale;
3. it can be compared across overlaps through shared observables or recoverable summaries.

This prevents the word "record" from collapsing into an arbitrary entangled correlation. The role of the record layer is to produce stable, repeatedly accessible data.

7.2 Working observer criterion

The OPH manuscript uses the tuple

$$O = (P_O, \mathcal{A}(P_O), \rho_O, R_O) \tag{12}$$

for an observer patch, local algebra, local state, and records. For the reference microphysics we add an explicit update interface and treat an observer as a pattern class rather than as a primitive object:

$$O_{\mathrm{ref}} = (P_O, \mathcal{A}(P_O), \rho_O, R_O, \mathcal{U}_O). \tag{13}$$

Here $\mathcal{U}_O$ is a local channel or update family that writes, compares, and repairs records using only observables available in P_O and on its overlaps.

A subsystem or pattern counts as an *observer in the present operational sense* if it satisfies all of the following.

1. **Patch access.** It is supported on a connected patch P_O with a nontrivial shared overlap algebra to neighboring patches.
2. **Record support.** It carries a set R_O of metastable record observables.
3. **Update capability.** It can condition future local actions on those records.
4. **Comparison capability.** It can query at least one shared overlap observable family and detect mismatch syndromes with neighbors.
5. **Persistence.** The pattern is identifiable for many local update cycles.

This is an observer pattern criterion, not a consciousness criterion. It is meant to support simulator definitions of record-bearing internal agents or agent-like subsystems.

```
Local registers --> records --> observer pattern --> synchronization

 link qudits / qubits
 |
 v
 gauge-invariant patch observables
 |
 v
 robust pointer variables / records
 |
 v
 persistent update-capable pattern O_ref
 |
 v
 compare shared overlap data and repair mismatch
```

8 Interaction, Synchronization, and Repair

The fixed-cutoff object used here is not a vague repair option or an abstract global repair law. It is a finite overlap interface that a simulator can instantiate on every edge of the patch-overlap graph.

For each neighboring patch pair (P, Q), the paper specifies the support of the update, the declared shared observable family, the finite syndrome alphabet, the local correction menu, the acceptance rule, and the logging contract. It does *not* claim a theorem that a chosen family must terminate, be confluent, or yield a refinement-consistent normal form. Those theorem-level questions are on the consensus-paper side of the boundary.

The default should be explicit syndrome-based overlap repair rather than a fully implicit dissipative dynamics. Dissipative or record-mediated variants may later be useful, but they should compile to the same fixed-cutoff interface: read shared data, compare it, choose one local candidate correction, accept or reject it by a declared contract, then refresh the record summaries and log the result.

8.1 The fixed-cutoff regulator and its finite interface layer

Fix a finite patch cover

$$\mathcal{P} = \{P_I\}_{I\in\mathcal{V}} \tag{14}$$

of the cellulated screen, and let

$$\mathcal{G}_{\text{patch}} = (\mathcal{V}, \mathcal{E}), \qquad \mathcal{E} = \{\{I, J\} : B_{IJ} := P_I \cap P_J \neq \varnothing\} \tag{15}$$

be the corresponding patch-overlap graph. For each patch P_I and overlap collar B_{IJ}, let $\widetilde{\mathcal{H}}_{P_I}$ and $\widetilde{\mathcal{H}}_{B_{IJ}}$ denote the finite tensor products of all link registers, auxiliary cut registers, and declared record registers supported on that patch or collar. Their regulator algebras are the boundary-fixed algebras

$$\mathcal{A}_I := \mathcal{B}(\widetilde{\mathcal{H}}_{P_I})^{G_{\partial P_I}}, \qquad \mathcal{A}_{IJ} := \mathcal{B}(\widetilde{\mathcal{H}}_{B_{IJ}})^{G_{\partial B_{IJ}}}. \tag{16}$$

Because the boundary gauge action on a patch restricts to the action on an overlap collar, there are canonical inclusion maps

$$\iota_{IJ\to I}(X) = X \otimes \mathbf{1}_{P_I \setminus B_{IJ}}, \qquad \iota_{IJ\to J}(X) = X \otimes \mathbf{1}_{P_J \setminus B_{IJ}}, \qquad X \in \mathcal{A}_{IJ}, \tag{17}$$

understood with identity on all omitted tensor factors.

At fixed cutoff, each collar algebra carries the OPH regulator decomposition

$$\mathcal{H}_{B_{IJ}} \cong \bigoplus_{\alpha\in\mathsf{A}_{IJ}} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}), \qquad Z(\mathcal{A}_{IJ}) = \bigoplus_{\alpha\in\mathsf{A}_{IJ}} \mathbb{C}\,\Pi_\alpha^{(IJ)}, \tag{18}$$

so the overlap center projectors are explicit regulator observables rather than placeholders. The same cutoff data also carry the finite patch-net interface later used on the consensus side.

Define the regulated microscopic package by

$$\mathfrak{R}_\Gamma = \begin{pmatrix} \Gamma,\ \mathcal{G}_{\text{patch}},\ \mathcal{H}_{\text{phys}},\ \{\mathcal{A}_I\}_I, \\ \{\mathcal{A}_{IJ}\}_{\{I,J\}\in\mathcal{E}},\ \{\iota_{IJ\to I}, \iota_{IJ\to J}\}_{\{I,J\}\in\mathcal{E}}, \\ \{\mathcal{O}_{IJ}^{\text{sh}}\}_{\{I,J\}\in\mathcal{E}},\ \{\text{read}_{I\to J}, \text{read}_{J\to I}\}_{\{I,J\}\in\mathcal{E}}, \\ \{R_I\}_I \end{pmatrix}. \tag{19}$$

The finite patch-net layer carried by that regulator is then

$$\mathfrak{F}_\Gamma = \Big(\mathcal{G}_{\text{patch}}, \{\mathsf{X}_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \{\mathbf{q}_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \{S_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \{L_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \{\Xi_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \{\mathcal{C}_{IJ}^{\min}\}_{\{I,J\}\in\mathcal{E}}\Big), \tag{20}$$

where $\mathbf{q}_{IJ}$ is the joint packet readout defined below. So the finite patch-net variables are not new primitives layered on top of OPH. They are the declared finite interface surface of the regulator itself.

8.2 Minimal overlap API

For each neighboring pair (P, Q) with shared collar $B_{PQ} = P \cap Q$, define the one-step update neighborhood N_{PQ} to be B_{PQ} together with all links, vertices, record registers, and faces at graph distance at most one from B_{PQ}. No single repair step is allowed to act outside N_{PQ}. A reference simulator should instantiate one typed overlap object

$$\mathfrak{I}_{PQ} = \Big(B_{PQ}, N_{PQ}, \mathcal{A}_{PQ}, \mathcal{O}^{\mathrm{sh}}_{PQ}, \mathsf{X}_{PQ}, \mathrm{read}_{P\to Q}, \mathrm{read}_{Q\to P}, \mathrm{compare}_{PQ}, \mathcal{C}^{\min}_{PQ}, \mathrm{accept}_{PQ}, \mathrm{commit}_{PQ}\Big) \tag{21}$$

for every edge of the patch-overlap graph. In the explicit builds below, the generic $\mathcal{C}^{\min}_{PQ}$ becomes the concrete abelian menu $\mathcal{C}^{(2)}_{IJ}$ or the nonabelian menu $\mathcal{C}^{(S_3)}_{IJ}$.

Its entries should be interpreted literally.

1. B_{PQ} is the collar support on which the shared observables live.
2. N_{PQ} is the maximal support on which one local repair move may act.
3. $\mathcal{A}_{PQ}$ is the regulator overlap algebra on that collar.
4. $\mathcal{O}^{\mathrm{sh}}_{PQ}$ is the declared shared observable family to be synchronized on this overlap.
5. X_{PQ} is the finite alphabet of readout packets exported by each side.
6. $\mathrm{read}_{P\to Q}$ and $\mathrm{read}_{Q\to P}$ are the local readout channels that turn the current microscopic state into those packets.
7. $\mathrm{compare}_{PQ}$ maps two packets into a finite mismatch syndrome.
8. $\mathcal{C}^{\min}_{PQ}$ is the finite menu of allowed local corrections.
9. accept_{PQ} checks whether a candidate correction satisfies the repair contract.
10. commit_{PQ} applies the accepted move, refreshes the declared record summaries, and writes the audit log.

In code, this is the per-overlap declaration table. Once $\mathfrak{I}_{PQ}$ has been fixed, different schedules may choose different overlaps or different candidate-order policies, but they should not silently change the shared observable family, the syndrome alphabet, or the acceptance rule mid-run.

8.3 Declared shared observable family and readout packets

What is synchronized is not the full microscopic quantum state. The synchronization target is a small declared family of overlap observables,

$$\mathcal{O}^{\mathrm{sh}}_{PQ} = \{\Pi^{(PQ)}_{\alpha}\}_{\alpha\in\mathsf{A}_{PQ}} \cup \{Q^{(PQ)}_{a}\}_{a=1}^{n_Q} \cup \{M^{(PQ)}_{r}\}_{r=1}^{n_M} \subseteq \mathcal{A}_{PQ}. \tag{22}$$

For phase 1, every overlap should expose at least one observable from each of the following three blocks.

1. **Sector block.** $\{\Pi^{(PQ)}_{\alpha}\}$ is a complete family of collar-sector projectors, with

$$\Pi^{(PQ)}_{\alpha}\Pi^{(PQ)}_{\beta} = \delta_{\alpha\beta}\Pi^{(PQ)}_{\alpha}, \qquad \sum_{\alpha\in\mathsf{A}_{PQ}} \Pi^{(PQ)}_{\alpha} = \mathbf{1}. \tag{23}$$

The inferred sector label α is the first overlap-level datum to compare.

2. **Boundary observable block.** Each $Q_a^{(PQ)}$ is a gauge-invariant collar observable with finite value alphabet V_a and declared comparison rule

$$\mathrm{diff}_a : \mathsf{V}_a \times \mathsf{V}_a \to \Delta_a. \tag{24}$$

For abelian pilots diff_a may simply be subtraction mod N or a binary equality test. For nonabelian pilots it may return a class, irrep, or other finite comparison label fixed in advance.

3. **Record-summary block.** Each $M_r^{(PQ)}$ is a read-stable summary variable built from the local record registers and/or recoverable overlap data, with finite value alphabet W_r, declared comparison rule

$$\mathrm{diff}_r : \mathsf{W}_r \times \mathsf{W}_r \to \Delta_r, \tag{25}$$

and freshness threshold $T_r^{\max}$. In the first simulator build these should be chosen from commuting or numerically stable approximately commuting pointer variables.

The readout channel from the P side of the overlap should return the packet

$$x_{P\to Q}^{(PQ)} = (\hat{g}_P, \hat{\alpha}_P, \hat{q}_{P,1}, \ldots, \hat{q}_{P,n_Q}, \hat{m}_{P,1}, \ldots, \hat{m}_{P,n_M}, \hat{s}_{P,1}, \ldots, \hat{s}_{P,n_M}) \in \mathsf{X}_{PQ}, \tag{26}$$

and similarly for $x_{Q\to P}^{(PQ)}$. Here:

1. $\hat{g}_P \in \{0,1\}$ is the local admissibility flag on the support $N_{PQ} \cap P$;
2. $\hat{\alpha}_P \in \mathsf{A}_{PQ}$ is the inferred collar-sector label;
3. $\hat{q}_{P,a} \in \mathsf{V}_a$ are the declared boundary-observable values;
4. $\hat{m}_{P,r} \in \mathsf{W}_r$ are the declared record-summary values;
5. $\hat{s}_{P,r} \in \{0,1\}$ is the stale-bit flag for record summary r, with $\hat{s}_{P,r} = 1$ meaning either that the age of the stored summary exceeds $T_r^{\max}$ or that the last refresh attempt failed.

A simulator build should treat the packet format as part of the API, not as an informal comment. Every overlap should declare the alphabets A_{PQ}, V_a, W_r, the comparison rules diff_a, diff_r, and the freshness thresholds before the first dynamics run starts.

8.4 Mismatch syndrome and local acceptance score

The mismatch syndrome is the finite object returned by the comparison map

$$S_{PQ} = \Big(\chi_G, \chi_\alpha, \{\Delta_a^Q\}_{a=1}^{n_Q}, \{\Delta_r^M\}_{r=1}^{n_M}, \{\sigma_r^{\mathrm{stale}}\}_{r=1}^{n_M}\Big), \tag{27}$$

$$\chi_G = 1 - \hat{g}_P \hat{g}_Q, \tag{28}$$

$$\chi_\alpha = \mathbf{1}[\hat{\alpha}_P \neq \hat{\alpha}_Q], \tag{29}$$

$$\Delta_a^Q = \mathrm{diff}_a(\hat{q}_{P,a}, \hat{q}_{Q,a}), \tag{30}$$

$$\Delta_r^M = \mathrm{diff}_r(\hat{m}_{P,r}, \hat{m}_{Q,r}), \tag{31}$$

$$\sigma_r^{\mathrm{stale}} = \hat{s}_{P,r} \vee \hat{s}_{Q,r}. \tag{32}$$

This definition makes the syndrome alphabet completely explicit: one gauge flag, one sector-mismatch flag, one finite comparison output for each declared boundary observable, one finite comparison output for each record summary, and one stale-bit flag per record summary.

For local acceptance it is convenient to use an exact lexicographic score rather than a single weighted scalar:

$$L_{PQ}(S_{PQ}) = \Big(\chi_G, \chi_\alpha, \sum_{a=1}^{n_Q} \mathbf{1}[\Delta_a^Q \neq 0], \sum_{r=1}^{n_M} \mathbf{1}[\Delta_r^M \neq 0], \sum_{r=1}^{n_M} \sigma_r^{\mathrm{stale}}\Big), \tag{33}$$

ordered lexicographically from left to right. An accepted correction should strictly lower L_{PQ}. This hard-codes the priority order

$$\begin{aligned} \text{gauge admissibility} &> \text{sector agreement} \\ &> \text{boundary-observable agreement} \\ &> \text{record-summary agreement} \\ &> \text{record freshness.} \end{aligned} \tag{34}$$

For logging and schedule diagnostics, it is useful to keep a scalar local cost,

$$\Xi_{PQ} = \lambda_G \chi_G + \lambda_\alpha \chi_\alpha + \sum_{a=1}^{n_Q} \lambda_a \, \mathbf{1}[\Delta_a^Q \neq 0] + \sum_{r=1}^{n_M} \lambda_r \, \mathbf{1}[\Delta_r^M \neq 0] + \lambda_{\mathrm{stale}} \sum_{r=1}^{n_M} \sigma_r^{\mathrm{stale}}, \qquad \lambda_\bullet \geq 0, \tag{35}$$

and the corresponding global mismatch potential

$$\Phi = \sum_{\langle P,Q \rangle} w_{PQ} \, \Xi_{PQ}. \tag{36}$$

The intended use is operational: L_{PQ} determines whether a single move is accepted, while Ξ_{PQ} and Φ are diagnostic quantities logged over a run. Nothing in this paper proves that a chosen schedule must decrease Φ globally.

8.5 Repair contract and minimal correction menu

The minimal correction menu should be intentionally small:

$$\mathcal{C}_{PQ}^{\min} = \mathcal{C}_{PQ}^{\mathrm{link}} \cup \mathcal{C}_{PQ}^{\mathrm{face}} \cup \mathcal{C}_{PQ}^{\mathrm{rec}} \cup \{\mathrm{noop}\}, \tag{37}$$

$$\mathcal{C}_{PQ}^{\mathrm{link}} = \{C_e^{(u)} : e \subset N_{PQ} \cap \partial B_{PQ}, \ u \in U_e\}, \tag{38}$$

$$\mathcal{C}_{PQ}^{\mathrm{face}} = \{C_f^{(u)} : f \cap B_{PQ} \neq \varnothing, \ u \in U_f\}, \tag{39}$$

$$\mathcal{C}_{PQ}^{\mathrm{rec}} = \{C_r^{\mathrm{refresh}} : r \text{ supported on } N_{PQ}\}. \tag{40}$$

Here U_e and U_f are finite move sets fixed by the microscopic model. For the octahedral $\mathbb{Z}_2$ pilot, a natural first choice is

$$U_e = \{X_e\}, \qquad U_f = \Big\{\prod_{e \subset \partial f} X_e\Big\}. \tag{41}$$

For a small nonabelian finite-group pilot, U_e may be left or right multiplication by a chosen generator set, and U_f the corresponding minimal plaquette moves. The record-refresh move C_r^{refresh} is allowed to rewrite only the declared summary variable and its stale bit from the current post-update overlap data; it is not allowed to invent a new synchronization target.

A candidate correction $c \in \mathcal{C}_{PQ}^{\min}$ should satisfy the following repair contract.

1. **Support contract.** $\mathrm{supp}(c) \subseteq N_{PQ}$.

2. **Gauge contract.** After tentative application of c, all Gauss checks on N_{PQ} must pass before the move can be committed.
3. **Shared-family contract.** The move is evaluated only against the predeclared family $\mathcal{O}_{PQ}^{\mathrm{sh}}$; the simulator may not enlarge or redefine that family during a run.
4. **Monotone local-fit contract.** If S_{PQ}^{c} is the post-move syndrome, then the move may be accepted only if
$$L_{PQ}(S_{PQ}^{c}) <_{\mathrm{lex}} L_{PQ}(S_{PQ}). \tag{42}$$
5. **Record contract.** If the move touches a declared record summary, the committed record value must match the post-update value of the corresponding $M_r^{(PQ)}$, and any stale bit that has actually been refreshed must be cleared.
6. **Audit contract.** Every attempted move must log the pre-packets, post-packets, syndrome, candidate identifier, support, and accept/reject reason.

A deterministic candidate selector is then

$$c_* = \arg\min_{c\in\mathcal{C}_{PQ}^{\min}} \Big(L_{PQ}(S_{PQ}^{c}), |\operatorname{supp}(c)|, \mathrm{id}(c)\Big), \tag{43}$$

where $\mathrm{id}(c)$ is a fixed total ordering on the candidate list and the minimum is taken only over candidates satisfying the support and gauge contracts. The move is committed only if it also satisfies the monotone local-fit contract. The no-op move is accepted only when $S_{PQ} = 0$; otherwise it is logged as an unresolved mismatch, not disguised as success.

8.6 Local update loop

Once every overlap object $\mathfrak{I}_{PQ}$ has been declared, the asynchronous local update loop should be completely explicit.

```
Initialize the microscopic state, the patch cover, the overlap objects I_PQ,
and the record registers.

For t = 1,...,T_max:
 choose one overlap (P,Q) according to schedule nu_t
 x_P <- read_{P->Q}^{(PQ)}
 x_Q <- read_{Q->P}^{(PQ)}
 S <- compare_{PQ}(x_P, x_Q)

 if S = 0:
 log no-op, Xi_{PQ}, and Phi
 continue

 build all trial candidates c in C_{PQ}^{min} \ {noop}
 discard candidates that violate local Gauss admissibility
 score each surviving candidate by
 (post-update L_{PQ}, support size, static candidate id)

 if no surviving candidate strictly lowers L_{PQ}:
```

```
log unresolved syndrome and continue

commit the best candidate
refresh every touched record summary and stale bit
reread x'_P, x'_Q and recompute S'_{PQ}, Xi_{PQ}, and Phi
log the accepted move
```

Only the schedule ν_t should vary across the simplest simulation experiments. Reasonable schedule choices are round-robin over overlap edges, uniform random asynchronous sampling, or priority by current largest Ξ_{PQ}. The interface $\mathfrak{I}_{PQ}$ should be unchanged across those schedule experiments.

This loop also makes the simulator/logging boundary concrete. A single log line per attempted overlap step should at minimum contain the overlap label (P, Q), the two input packets, the syndrome, the chosen candidate (or explicit failure), the post-update packet values, the local score L_{PQ} before and after, and the diagnostics Ξ_{PQ} and Φ. Without that log, later schedule and obstruction analysis is underdetermined.

8.7 Local repair instrument and the packet-closure boundary

For a fixed backend, the overlap repair step is the staged instrument

$$\mathcal{R}_{PQ} := \mathcal{W}_{PQ} \circ \mathcal{C}_{PQ} \circ \mathcal{M}_{PQ} \circ \mathcal{P}_{PQ}, \tag{44}$$

where

$$\mathcal{P}_{PQ} := \left(\mathrm{read}_{P\to Q}, \mathrm{read}_{Q\to P}\right), \qquad \mathcal{M}_{PQ} := \mathrm{compare}_{PQ}, \tag{45}$$

$\mathcal{C}_{PQ}$ is the candidate-selection and acceptance stage based on the declared menu and repair contract, and $\mathcal{W}_{PQ} := \mathrm{commit}_{PQ}$ performs the accepted local move together with the permitted record refresh. Every committed branch is supported on N_{PQ} and acts as the identity outside that neighborhood.

This gives the object-level embedding used in the regulator lane: packets, syndromes, scores, candidate menus, and repair branches are all regulator-side data. A closed autonomous dynamics on packet space does not follow automatically. Define the joint packet readout

$$\mathbf{q}_{PQ}(\rho) := \left(\mathrm{read}_{P\to Q}(\rho), \mathrm{read}_{Q\to P}(\rho)\right), \tag{46}$$

and the packet-equivalence relation

$$\rho \sim_{PQ} \rho' \quad \Longleftrightarrow \quad \mathbf{q}_{PQ}(\rho) = \mathbf{q}_{PQ}(\rho'). \tag{47}$$

The finite packet layer is autonomously closed on overlap (P, Q) only if, for every accepted candidate branch c,

$$\mathbf{q}_{PQ}\left(\mathcal{R}^c_{PQ}(\rho)\right) = \mathbf{q}_{PQ}\left(\mathcal{R}^c_{PQ}(\rho')\right) \qquad \text{whenever } \rho \sim_{PQ} \rho'. \tag{48}$$

If this packet-closure test passes, the microscopic instrument pushes forward to a genuine finite update rule on packet space. If it fails, the finite packet layer is an embedded readout/actuation interface, without an autonomous quotient dynamics. This is the boundary kept explicit for the later fixed-point paper *Reality as a Consensus Protocol*.

8.8 Acceptance Tests and Handoff Boundary

The first simulator build should be required to pass explicit interface-level tests before any larger interpretation is attached to it.

1. **Schema completeness.** Every overlap in the patch-overlap graph must ship with declared alphabets A_{PQ}, V_a, W_r, comparison rules diff_a, diff_r, freshness thresholds $T_r^{\max}$, and a finite correction menu. Missing declarations are build failures.

2. **Zero-syndrome sanity.** If two overlap packets agree field by field and all stale bits are zero, then $\mathrm{compare}_{PQ}$ must return $S_{PQ} = 0$ and the only legal response is the no-op.

3. **Single-defect detectability.** A manually injected mismatch in the sector label, one boundary observable, or one record summary must flip the corresponding syndrome component and should not generate unrelated syndrome components except through an explicit gauge violation.

4. **Gauge-safe commit.** Every accepted nontrivial correction must leave all Gauss checks on N_{PQ} satisfied.

5. **Strict local improvement.** For every accepted correction,

$$L_{PQ}^{\mathrm{after}} <_{\mathrm{lex}} L_{PQ}^{\mathrm{before}}. \tag{49}$$

   Reject logs must state which repair-contract clause failed.

6. **Record-refresh fidelity.** After an accepted refresh move, rereading the overlap must return record-summary values equal to the current declared $M_r^{(PQ)}$ values, and the corresponding stale bits must clear.

7. **Packet-closure exposure.** For each overlap and each accepted candidate branch, the exact backend should test whether two microscopic states with the same input packet pair can produce different reread packets on the declared shared family. If that happens, the run must log the hidden-state dependence explicitly, and the build must not claim an autonomous packet-only dynamics for that observable family.

8. **Schedule comparison on a small cover.** On the octahedral $\mathbb{Z}_2$ pilot with a three-patch tree-like cover, rerun the same initial condition under round-robin, random, and largest-Ξ schedules and compare the final shared-observable packets. Agreement on that declared family is evidence that the interface is functioning on that instance; disagreement is a measured failure mode, not a contradiction of any theorem in this paper.

9. **Residual-obstruction exposure.** On a deliberately frustrated cycle, the loop should surface unresolved syndromes or schedule dependence in the logs rather than silently overwriting record summaries to fake agreement.

This paper owns the fixed-cutoff regulator $\mathfrak{R}_\Gamma$, the embedded finite interface layer $\mathfrak{F}_\Gamma$, the overlap object $\mathfrak{I}_{PQ}$, the local repair instrument $\mathcal{R}_{PQ}$, the packet-closure tests, and the unit/integration tests that a simulator can run. *Reality as a Consensus Protocol* owns the next layer: autonomous packet dynamics when closure holds, repair completeness on the packet-closed quotient branch, the quotient-level compatibility package that yields the local diamond, the resulting schedule-independent normal-form theory, and the classification of residual holonomy or obstruction classes. This paper therefore supplies the concrete microscopic inputs to the consensus story without claiming that the consensus theorems are proved on this microphysics surface.

9 What a First Implementable Reference Model Should Contain

A first actual build should be deliberately small, explicit, and reproducible. The target of the first round is not a unique UV claim; it is one finite cutoff model with enough declared structure that two independent teams would compile the same registers, the same layers, the same logged observables, and the same validation tests. It should carry two coupled deliverables on the same microscopic screen: the record/repair interface of the synchronization architecture, and an explicit edge-sector thermal branch on the same declared overlaps so the edge heat-kernel/Casimir law is tested on the same regulator data rather than on a separate toy model.

9.1 Fixed geometry and patch cover

Use the octahedral cellulation of S^2. Label vertices

$$X^{\pm}, \qquad Y^{\pm}, \qquad Z^{\pm}, \tag{50}$$

and faces by sign triples

$$f_{\sigma\tau\upsilon} = (X^{\sigma}, Y^{\tau}, Z^{\upsilon}), \qquad \sigma, \tau, \upsilon \in \{+, -\}. \tag{51}$$

Thus $|V| = 6$, $|E| = 12$, and $|F| = 8$. Use the fixed vertex order

$$X^+ \prec X^- \prec Y^+ \prec Y^- \prec Z^+ \prec Z^-, \tag{52}$$

store one physical link register on each undirected edge, orient that edge from the earlier endpoint to the later endpoint, and use the same ordered endpoint pair both for register lookup and for the lexicographic ordering of collar edges in the repair layer. Every holonomy routine must use that same orientation and invert the group element when the traversal direction is reversed.

For the octahedral pilot, use the patch cover

$$P_X = \{f_{+\tau\upsilon} : \tau, \upsilon \in \{+, -\}\}, \tag{53}$$
$$P_Y = \{f_{\sigma+\upsilon} : \sigma, \upsilon \in \{+, -\}\}, \tag{54}$$
$$P_Z = \{f_{\sigma\tau+} : \sigma, \tau \in \{+, -\}\}. \tag{55}$$

The pairwise overlaps are

$$B_{XY} = P_X \cap P_Y = \{f_{+++}, f_{++-}\}, \tag{56}$$
$$B_{XZ} = P_X \cap P_Z = \{f_{+++}, f_{+-+}\}, \tag{57}$$
$$B_{YZ} = P_Y \cap P_Z = \{f_{+++}, f_{-++}\}, \tag{58}$$

and the triple overlap is the single face f_{+++}. This specific cover is the required reference-model geometry because it contains pairwise collars, a nontrivial triple-overlap consistency check, and one face unique to each patch:

$$f_X^{\text{bulk}} = f_{+--}, \qquad f_Y^{\text{bulk}} = f_{-+-}, \qquad f_Z^{\text{bulk}} = f_{--+}. \tag{59}$$

For the later compare and repair layers, also fix

$$(f_{XY}^+, f_{XY}^-) = (f_{+++}, f_{++-}), \tag{60}$$
$$(f_{XZ}^+, f_{XZ}^-) = (f_{+++}, f_{+-+}), \tag{61}$$
$$(f_{YZ}^+, f_{YZ}^-) = (f_{+++}, f_{-++}). \tag{62}$$

The same cover is used both for synchronization and for edge-law readout. No separate "edge-only" geometry is introduced.

9.2 First build: the $\mathbb{Z}_2$ pilot

The first complete build should use $G = \mathbb{Z}_2$ and a stabilizer-friendly discrete update rule. Put one qubit q_e on each of the 12 stored links, with

$$|0\rangle_e,\ |1\rangle_e \tag{63}$$

the Z_e-eigenbasis. Use

$$A_v = \prod_{e \ni v} X_e, \qquad P_v = \frac{1 + A_v}{2}, \qquad B_f = \prod_{e \subset \partial f} Z_e. \tag{64}$$

For the reference pilot, the record layer uses three qubits per patch,

$$R_X = (r_X^{\text{bulk}}, r_X^{(Y)}, r_X^{(Z)}), \quad R_Y = (r_Y^{\text{bulk}}, r_Y^{(X)}, r_Y^{(Z)}), \quad R_Z = (r_Z^{\text{bulk}}, r_Z^{(X)}, r_Z^{(Y)}), \tag{65}$$

for a total of 9 record qubits. Their declared meanings are

$$Z_{r_I^{\text{bulk}}} \leftrightarrow B_{f_I^{\text{bulk}}}, \tag{66}$$

$$Z_{r_I^{(J)}} \leftrightarrow C_{IJ} := \prod_{e \in \partial B_{IJ}} Z_e, \tag{67}$$

where C_{IJ} is the four-edge collar-loop parity around the two-face overlap B_{IJ}.

The required initial state is any explicitly prepared state in the Gauss-invariant subspace. The default debug initialization is

$$|\Omega_0\rangle \propto \Big(\prod_{v \in V} P_v\Big)\Big(\prod_{f \in F} \frac{1 + B_f}{2}\Big)|+\rangle^{\otimes 12} \otimes |0\rangle_{\text{rec}}^{\otimes 9}, \tag{68}$$

followed by one record-write layer so that all record qubits contain the declared summaries of the initialized link state.

9.3 Required local layers for the $\mathbb{Z}_2$ pilot

The first build should be a layered discrete-time simulator. One full cycle applies the layers in the order

$$L_{\text{bulk}} \longrightarrow L_{\text{check}} \longrightarrow L_{\text{write}} \longrightarrow L_{\text{compare}} \longrightarrow L_{\text{repair}} \longrightarrow L_{\text{verify}}. \tag{69}$$

The layers are fixed as follows.

1. **Bulk face-update layer L_{bulk}.** On a chosen active face set $F_{\text{act}}(t)$, apply either the identity or the face move

$$F_f := \prod_{e \subset \partial f} X_e. \tag{70}$$

   The default schedule is a deterministic scan over the eight faces with one Bernoulli face-move attempt per face and default attempt rate $p_{\text{face}} = 0.1$.

2. **Gauge-check layer L_{check}.** Measure or compute all six star values

$$a_v(t) \in \{+1, -1\}, \qquad a_v(t) = \langle A_v \rangle_t \tag{71}$$

   in the exact backend, or their sampled eigenvalues in the shot backend. Because the allowed reference-model bulk and repair moves all commute with every A_v, any nonzero star syndrome in the pilot is treated as an implementation fault, not as physical dynamics.

3. **Record-write layer** L_{write}**.** Overwrite the record qubits with the declared summaries

$$r_I^{\text{bulk}} \leftarrow B_{f_I^{\text{bulk}}}, \tag{72}$$

$$r_I^{(J)} \leftarrow C_{IJ}, \tag{73}$$

using parity-compute and uncompute subcircuits or their exact-state equivalents. The required mode for the first build is overwrite, not additive update, so that stale-record diagnostics are unambiguous.

4. **Overlap-compare layer** L_{compare}**.** Choose one active overlap from

$$(XY), (XZ), (YZ) \tag{74}$$

by cyclic schedule or by a logged pseudorandom seed. Measure the live shared observables on that overlap:

$$c_{IJ} = C_{IJ}, \qquad b_{IJ}^+ = B_{f_{IJ}^+}, \qquad b_{IJ}^- = B_{f_{IJ}^-}, \tag{75}$$

together with the two cached record values

$$m_I^{(J)} = Z_{r_I^{(J)}}, \qquad m_J^{(I)} = Z_{r_J^{(I)}}. \tag{76}$$

For the named overlaps the face labels are

$$(f_{XY}^+, f_{XY}^-) = (f_{+++}, f_{++-}), \tag{77}$$

$$(f_{XZ}^+, f_{XZ}^-) = (f_{+++}, f_{+-+}), \tag{78}$$

$$(f_{YZ}^+, f_{YZ}^-) = (f_{+++}, f_{-++}). \tag{79}$$

5. **Repair layer** L_{repair}**.** Evaluate the explicit candidate menu

$$\mathcal{C}_{IJ}^{(2)} = \{\text{Id}, \ \text{Refresh}_I, \ \text{Refresh}_J, \ \text{Refresh}_{IJ}\} \cup \{X_e : e \in \partial B_{IJ}\} \cup \{F_f : f \subset B_{IJ}\}. \tag{80}$$

For each candidate $C \in \mathcal{C}_{IJ}^{(2)}$, compute the post-correction local cost

$$\Xi_{IJ}^{(2)}(C) = \lambda_c \mathbf{1}[m_I^{(J)} \neq c_{IJ}]_C + \lambda_c \mathbf{1}[m_J^{(I)} \neq c_{IJ}]_C + \lambda_r \mathbf{1}[m_I^{(J)} \neq m_J^{(I)}]_C, \tag{81}$$

with default weights

$$\lambda_c = 2, \qquad \lambda_r = 1. \tag{82}$$

Choose the unique minimizer if one exists. If there is a tie, use the fixed tie-break order

$$\text{Refresh}_I \prec \text{Refresh}_J \prec \text{Refresh}_{IJ} \prec X_{e_1} \prec X_{e_2} \prec X_{e_3} \prec X_{e_4} \prec F_{f_{IJ}^+} \prec F_{f_{IJ}^-} \prec \text{Id}, \tag{83}$$

where $e_1, \ldots, e_4$ is the lexicographic ordering of the four collar edges induced by the stored endpoint order above. This tie-break order must be logged and fixed across all runs.

6. **Verification layer** L_{verify}**.** Recompute a_v, the active overlap data, and the global mismatch potential

$$\Phi(t) = \Xi_{XY}^{(2)}(t) + \Xi_{XZ}^{(2)}(t) + \Xi_{YZ}^{(2)}(t). \tag{84}$$

A cycle is accepted only after the post-repair measurements have been written to the log.

The designated observer proxy for the pilot is patch P_X: its controller is required to choose the active correction on B_{XY} or B_{XZ} using only the locally available record bits in R_X plus the declared overlap data. That is enough to instantiate the paper's operational observer criterion at fixed cutoff without making stronger claims.

9.4 Declared edge-law branch on the same overlaps

The heat-kernel measurement is not taken on the record bit C_{IJ} alone. Each overlap must additionally declare one electric-center cut register h_{IJ} extracted from the same collar B_{IJ}. At fixed cutoff this is the boundary register that carries the boundary gauge action used in the edge-center completion. On the abelian pilot it is a single coarse-grained $\mathbb{Z}_2$ cut bit; on the nonabelian pilot it is the encoded S_3 cut register; on the quantum-link upgrade it is the truncated irrep-label register.

The required declared data on every overlap are

$$\Big(h_{IJ}, \{\Pi_R^{(IJ)}\}_R, \Delta_{IJ}^{\text{edge}}, t_{IJ}\Big), \tag{85}$$

where h_{IJ} is the cut register, $\Pi_R^{(IJ)}$ are its sector projectors, $\Delta_{IJ}^{\text{edge}}$ is the class-averaged edge Laplacian of Section 12.2, and t_{IJ} is the overlap inverse-temperature parameter or temperature grid used in a benchmark run.

Two execution modes are required.

1. **Exact edge-state mode.** On one chosen overlap, prepare the fixed-cutoff reduced edge state

$$\rho_{IJ}^{\text{edge}}(t) = \bigoplus_R \frac{d_R e^{-t\lambda_R^{(IJ)}}}{Z_{IJ}(t)} \frac{\mathbf{1}_{V_R}}{d_R}, \qquad Z_{IJ}(t) = \sum_{R'} d_{R'} e^{-t\lambda_{R'}^{(IJ)}}, \tag{86}$$

or equivalently exponentiate the declared cut-register edge Hamiltonian before tracing one side, and then read out $\langle \Pi_R^{(IJ)} \rangle_t$.

2. **Local sampler mode.** Use the same inverse-closed collar move set S_{IJ} to run the explicit Metropolis kernel of Section 12.2 and compare long-run frequencies with the exact edge-state weights.

The repair menu and the thermal menu may differ term by term, but they must be drawn from the same gauge-preserving local screen operations on N_{IJ}. That is what makes the edge law a statement about the present microphysics rather than about a separate external model.

For the $\mathbb{Z}_2$ sanity build, the sector projectors on the coarse cut register are the two electric-center projectors

$$\Pi_0^{(IJ)} = \frac{1+Q_{IJ}}{2}, \qquad \Pi_1^{(IJ)} = \frac{1-Q_{IJ}}{2}, \tag{87}$$

where Q_{IJ} is the declared coarse electric-center bit on that overlap. Because $d_0 = d_1 = 1$, this build checks the abelian Boltzmann/heat-kernel shape but does not expose the nonabelian d_R factor.

9.5 First nonabelian upgrade: the S_3 build

The first upgrade should keep the same octahedral graph, the same patch cover, the same overlap schedule, and the same log schema. Only the link and record alphabets change. Each link register becomes a six-state S_3 register encoded into three qubits with the fixed lookup table

$$000 \leftrightarrow e, \qquad 001 \leftrightarrow (12), \qquad 010 \leftrightarrow (13), \tag{88}$$

$$011 \leftrightarrow (23), \qquad 100 \leftrightarrow (123), \qquad 101 \leftrightarrow (132), \tag{89}$$

while 110 and 111 are illegal code words. Reverse edge traversal uses the inverse group element, implemented by the fixed inversion lookup

$$e^{-1} = e, \qquad (12)^{-1} = (12), \qquad (13)^{-1} = (13), \qquad (23)^{-1} = (23), \qquad (123)^{-1} = (132). \tag{90}$$

At each vertex, the required Gauss projector is the exact finite-group projector

$$P_v^{S_3} = \frac{1}{6} \sum_{h \in S_3} U_v(h), \tag{91}$$

with $U_v(h)$ acting by left or right multiplication on the incident oriented links according to their incoming or outgoing orientation. For each face f, compute the ordered face holonomy

$$h_f = \prod_{e \subset \partial f} g_e^{\epsilon(e,f)}, \tag{92}$$

and for each overlap collar compute

$$h_{IJ}^{\text{collar}} = \prod_{e \in \partial B_{IJ}} g_e^{\epsilon(e,IJ)}. \tag{93}$$

The required record layer for the first S_3 build is binary-coded rather than one-hot. Each patch P_I carries

$$R_I^{S_3} = \left(c_I^{\text{bulk}}, c_I^{(J_1)}, c_I^{(J_2)}, v_I\right), \tag{94}$$

where each c-register is a two-qubit class label with encoding

$$00 \leftrightarrow [e], \qquad 01 \leftrightarrow [\tau], \qquad 10 \leftrightarrow [\sigma], \qquad 11 \text{ illegal}, \tag{95}$$

and v_I is a one-bit valid-write flag set to 1 only when the current write produced a legal class label. Here $[\tau]$ is the transposition class and $[\sigma]$ the three-cycle class.

The local layer structure is the same as in the $\mathbb{Z}_2$ pilot, but the candidate correction set is

$$\begin{aligned} \mathcal{C}_{IJ}^{(S_3)} =& \{\text{Id},\ \text{Refresh}_I,\ \text{Refresh}_J,\ \text{Refresh}_{IJ}\} \\ & \cup \{L_e(s), L_e(r), L_e(r^{-1}), R_e(s), R_e(r), R_e(r^{-1}) : e \in \partial B_{IJ}\} \\ & \cup \{M_f(s), M_f(r), M_f(r^{-1}) : f \subset B_{IJ}\}, \end{aligned} \tag{96}$$

where $s = (12)$, $r = (123)$, $L_e(g)$ and $R_e(g)$ left- or right-multiply one collar edge by a generator, and $M_f(g)$ multiplies one chosen face boundary by the same group element in the oriented order. The default nonabelian local cost is

$$\Xi_{IJ}^{(S_3)}(C) = \lambda_c\, \mathbf{1}[c_I^{(J)} \neq \text{cl}(h_{IJ}^{\text{collar}})]_C + \lambda_c\, \mathbf{1}[c_J^{(I)} \neq \text{cl}(h_{IJ}^{\text{collar}})]_C + \lambda_r\, \mathbf{1}[c_I^{(J)} \neq c_J^{(I)}]_C + \lambda_u\, N_{\text{illegal}}(C), \tag{97}$$

with defaults

$$\lambda_c = 2, \qquad \lambda_r = 1, \qquad \lambda_u = 10. \tag{98}$$

In addition to conjugacy-class data, the first S_3 upgrade must also log the three collar irrep projectors

$$\Pi_\rho^{(IJ)} = \frac{d_\rho}{6} \sum_{g \in S_3} \chi_\rho(g^{-1})\, U_g^{(IJ)}, \qquad \rho \in \{\mathbf{1}, \mathbf{1'}, \mathbf{2}\}, \tag{99}$$

because the whole point of the upgrade is to check that the overlap interface behaves well when the shared sector carries genuinely nonabelian labels. For the edge-law branch, the cut register on each active overlap is the same encoded S_3 boundary register, and the minimal class-symmetric edge operator is the transposition-class Laplacian

$$\Delta_{IJ}^{(S_3)} = 3\mathbf{1} - \sum_{\tau \in T} U_\tau^{(IJ)}, \qquad T = \{(12), (13), (23)\}. \tag{100}$$

Its sector eigenvalues are

$$\lambda_{\mathbf{1}} = 0, \qquad \lambda_{\mathbf{1}'} = 6, \qquad \lambda_{\mathbf{2}} = 3. \tag{101}$$

So the exact nonabelian edge law on one overlap is

$$p_{\mathbf{1}} : p_{\mathbf{1}'} : p_{\mathbf{2}} = 1 : e^{-6t} : 2e^{-3t}. \tag{102}$$

This build is the first one in which the d_R factor is visible rather than absorbed into equal abelian multiplicities. A correct simulator must therefore fit the $2e^{-3t}$ weight of the standard irrep better than the wrong law e^{-3t} that would arise if the entropic d_R factor were missing.

9.6 Compact-group truncation as the merge-boundary build

Once the finite-group branch passes, one overlap should be upgraded to the declared quantum-link style truncation. Choose a finite irrep set $\mathcal{R}_{\max}$ and encode each cut register as

$$|R, m, n\rangle \longmapsto |R\rangle_{\text{irrep}} \otimes |m\rangle_{\text{left}} \otimes |n\rangle_{\text{right}}. \tag{103}$$

On that truncated support, the required edge operator is

$$\Delta^{\text{QL}}_{IJ} = \sum_{R \in \mathcal{R}_{\max}} C_2(R)\, \Pi^{(IJ)}_R. \tag{104}$$

The exact fixed-cutoff sector law is then

$$p^{(IJ)}_R(t) = \frac{d_R e^{-tC_2(R)}}{\sum_{R' \in \mathcal{R}_{\max}} d_{R'} e^{-tC_2(R')}}. \tag{105}$$

This is enough for the SM/GR-paper merge boundary at fixed cutoff. What is *not* claimed here is the refinement-stable uniqueness of the truncation, the refinement-limit universality of t, or the theorem that different truncation ladders must land on one common compact limit. Those are later merge-level questions.

9.7 Required deliverables of the first build package

An implementation should treat the following outputs as mandatory rather than optional.

1. A machine-readable geometry file containing the vertex list, oriented edge list, face list, patch memberships, overlap memberships for the octahedron, and the designated cut register for each overlap.

2. A register file declaring every link register, every declared cut register used for sector readout, every record register, and every illegal-code word.

3. A layer file specifying $L_{\text{bulk}}, L_{\text{check}}, L_{\text{write}}, L_{\text{compare}}, L_{\text{repair}}, L_{\text{verify}}$, and any edge-thermal layer in the exact order used by the run.

4. A correction file containing the candidate menus $\mathcal{C}^{(2)}_{IJ}$ and $\mathcal{C}^{(S_3)}_{IJ}$, the weights $(\lambda_c, \lambda_r, \lambda_u)$, and the tie-break ordering.

5. An edge-law file containing, for every overlap, the declared sector projectors $\Pi^{(IJ)}_R$, the chosen edge operator $\Delta^{\text{edge}}_{IJ}$, its eigenvalues $\lambda^{(IJ)}_R$, the temperature grid used in validation, and the execution mode (exact edge-state or local sampler).

6. A log schema that writes raw register states, derived observables, selected corrections, sector probabilities, attempted edge moves, and pass/fail statistics for every cycle.

This is a fixed-cutoff microphysics note. A successful pilot would show that one fixed-cutoff screen model can instantiate the declared objects and measurements; it would not identify a unique UV completion or settle any refinement-limit claim.

10 Simulator Validation Package

A useful simulator package has to be able to fail. The validation target is not refinement-limit closure. It is the fixed-cutoff architecture claimed here: gauge-admissible local layers, measurable overlap observables, persistent records, and a repair loop whose behavior can be diagnosed. The validation package should therefore be distributed with the first build, not added later as an informal plotting appendix.

10.1 Required benchmark families

Every implementation must run the same named benchmark suite. The minimum required suite is:

1. `Z2-0-clean`: initialize the $\mathbb{Z}_2$ pilot in $|\Omega_0\rangle$, run 12 cycles, and inject no defect.
2. `Z2-1-stale-record`: after the first L_{write}, flip $r_X^{(Y)}$ and leave the link state untouched; run 12 cycles.
3. `Z2-2-collar-flip`: before the first L_{compare} on B_{XY}, apply X_e on the lexicographically first collar edge of B_{XY}; run 12 cycles.
4. `Z2-3-double-overlap`: inject one collar-edge flip on B_{XY} and one on B_{YZ} before the first compare layer; run 15 cycles.
5. `Z2-4-frustrated-cycle`: after every write layer during cycles 0 through 4, force $r_X^{(Y)}$ to the opposite of the live collar bit on B_{XY}; run 15 cycles to test persistent residual mismatch.
6. `S3-0-clean`: initialize the S_3 build in the declared identity sector, run 15 cycles, and inject no defect.
7. `S3-1-transposition`: before the first compare layer on B_{XY}, left-multiply the lexicographically first collar edge by (12); run 20 cycles.
8. `S3-2-threecycle`: before the first compare layer on B_{XY}, left-multiply the same edge by (123); run 20 cycles.
9. `S3-3-mixed-class-cycle`: inject a transposition-class defect on B_{XY} and a three-cycle defect on B_{YZ} before the first compare layer; run 20 cycles.

For each benchmark, run all six sweep orders of the three overlaps (XY, XZ, YZ). The analytic backend must run each order once exactly. The sampled backend must run each order with 32 independent seeds and 1024 shots per seed. A reported reference-model pass is not valid unless it includes all named benchmarks and all six schedule orders.

The canonical executable surface for the analytic fixed-cutoff suite is `code/consensus/reference_architecture_benchmark_suite.py`. Its current emitted artifact is `code/consensus/r` uns/reference_architecture_benchmark_suite_current.json. That artifact is the issue-237 reference-architecture benchmark surface: it runs the named $\mathbb{Z}_2$ and S_3 families over all six schedules, records packet-closure and repair-completeness gates, includes the residual-obstruction gate, and states explicitly that the result is fixed-cutoff only. It is not a refinement-limit lift and it is not a uniqueness theorem for microscopic UV completion.

10.2 Required run manifest and per-cycle log

Every run must write two artifacts.

Run manifest. The manifest is one record with the fields build_id, group, backend, encoding, graph_id, patch_cover_id, benchmark_id, schedule_id, seed, shots, cycles, candidate_menu_hash, and tie_break_hash. The hashes are required so that two runs can be compared without guessing whether the correction table changed.

Cycle table. Each cycle must write one row with the fields cycle, active_overlap, link_state_pre, record_state_pre, a_v_pre[6], face_obs_pre[8], collar_obs_pre[3], candidate_costs, chosen_correction, accepted, tie_break_rank, link_state_post, record_state_post, a_v_post[6], face_obs_post[8], collar_obs_post[3], Phi_post, g_post, and u_bad_post. For the $\mathbb{Z}_2$ pilot, link_state is a 12-bit string in stored-edge order and face_obs is the ordered list of eight plaquette bits B_f. For the S_3 build, link_state must contain both the raw three-bit code word and the decoded group symbol on every edge, while face_obs must contain the eight decoded face holonomies together with their conjugacy classes. For the S_3 build, collar_obs must additionally contain the three irrep-probability triples

$$(p_{\mathbf{1}}^{(IJ)}, p_{\mathbf{1'}}^{(IJ)}, p_{\mathbf{2}}^{(IJ)}). \tag{106}$$

The cycle table is not optional. A run that stores only aggregate plots is a failed validation artifact because it cannot support repair-level diagnosis.

10.3 Derived measured quantities

Let

$$c_{IJ}(t) = \begin{cases} C_{IJ}(t), & \mathbb{Z}_2, \\ \mathrm{cl}(h_{IJ}^{\mathrm{collar}}(t)), & S_3, \end{cases} \tag{107}$$

and let $m_I^{(J)}(t)$ be the corresponding cached overlap record. From the manifest and cycle table, compute at least the following quantities. Here $\Xi_{IJ}(t)$ denotes the build-appropriate local mismatch cost on overlap B_{IJ}, namely $\Xi^{(2)}_{IJ}(t)$ in the $\mathbb{Z}_2$ pilot and $\Xi^{(S_3)}_{IJ}(t)$ in the S_3 build:

$$g(t) = \sum_{v\in V} \mathbf{1}[a_v(t) = -1], \tag{108}$$

$$u_{\mathrm{bad}}(t) = N_{\text{illegal link codes}}(t) + N_{\text{illegal record codes}}(t), \tag{109}$$

$$\Phi(t) = \sum_{IJ\in\{XY,XZ,YZ\}} \Xi_{IJ}(t), \tag{110}$$

$$G_\Phi = \frac{1}{N_{\mathrm{acc}}} \sum_{t\in\text{accepted}} \big(\Phi(t-1) - \Phi(t)\big), \tag{111}$$

$$C_r(k) = \frac{1}{N_{\mathrm{rec}}} \sum_r \Pr[r(t_0) = r(t_0+k)], \tag{112}$$

$$Q_{\mathrm{readback}} = \max_{IJ} \|\mu^{\mathrm{read}}_{IJ} - \mu^{\mathrm{skip}}_{IJ}\|_1, \tag{113}$$

$$D_{\mathrm{sched}} = \max_{s,s'} \sum_{IJ} \|\mu^{(s)}_{IJ} - \mu^{(s')}_{IJ}\|_1, \tag{114}$$

$$M_{\mathrm{class}} = \frac{1}{N_{\mathrm{meas}}} \sum \mathbf{1}[\widehat{\mathrm{cl}}(h^{\mathrm{collar}}_{IJ}) \neq \mathrm{cl}(h^{\mathrm{collar}}_{IJ})], \tag{115}$$

$$N_{\mathrm{irrep}} = \max_{IJ,t} \left| p^{(IJ)}_{\mathbf{1}}(t) + p^{(IJ)}_{\mathbf{1}'}(t) + p^{(IJ)}_{\mathbf{2}}(t) - 1 \right|. \tag{116}$$

Here G_Φ is the mean accepted improvement in the mismatch potential, $C_r(k)$ is the record-retention curve, Q_{readback} measures back-action from an inserted record-read layer, D_{sched} measures schedule dependence on the declared shared observable family, M_{class} measures nonabelian class-label errors, and N_{irrep} checks whether the logged irrep probabilities even define a normalized decomposition.

In addition, every run must report the convergence time N_{conv}, defined for removable-defect benchmarks as the first cycle after which $\Phi = 0$ for the rest of the run, and the residual-obstruction plateau time N_{plat}, defined for frustrated benchmarks as the first cycle after which both the residual label and the residual value of Φ stop changing.

10.4 Pass and Fail Gates

The validation package should use explicit gates rather than qualitative language.

1. **Kinematic gate.** In the analytic backend, every post-verify cycle must satisfy

$$g(t) = 0, \qquad u_{\mathrm{bad}}(t) = 0. \tag{117}$$

   In the sampled backend, the required bounds are

$$\max_t \frac{g(t)}{|V|} \le 10^{-3}, \qquad \max_t \frac{u_{\mathrm{bad}}(t)}{N_{\mathrm{reg}}} \le 10^{-4}, \tag{118}$$

   where N_{reg} is the number of encoded link and record code words checked for legality in that build.

2. **Record gate.** On `Z2-0-clean`, `Z2-1-stale-record`, and `S3-0-clean`, the required record metrics are

$$C_r(5) \ge 0.80, \qquad Q_{\mathrm{readback}} \le 0.05. \tag{119}$$

   If these fail while the kinematic gate passes, the architecture has local observables while lacking a usable record layer.

3. $\mathbb{Z}_2$ **synchronization gate.** On `Z2-1-stale-record`, `Z2-2-collar-flip`, and `Z2-3-double-overlap`, the analytic backend must reach $\Phi = 0$ within 12 repair activations for all six schedules. The sampled backend must do so within 15 repair activations in at least 95% of runs. In both backends, one also requires

$$G_\Phi > 0, \qquad D_{\text{sched}} \le \begin{cases} 10^{-8}, & \text{analytic}, \\ 0.05, & \text{sampled}. \end{cases} \tag{120}$$

4. **Residual-obstruction gate.** On `Z2-4-frustrated-cycle`, the build is not supposed to fake full agreement. The required behavior is

$$g(t) = 0 \text{ for all post-verify cycles}, \qquad N_{\text{plat}} \le 8, \tag{121}$$

and the residual label must remain unchanged for the final 10 cycles. A run that oscillates indefinitely, or that reports $\Phi = 0$ while the forced stale source is present, fails this gate.

5. S_3 **nonabelian gate.** On `S3-1-transposition`, `S3-2-threecycle`, and `S3-3-mixed-class-cycle`, the required label-fidelity bounds are

$$M_{\text{class}} \le \begin{cases} 0, & \text{analytic}, \\ 0.05, & \text{sampled}, \end{cases} \qquad N_{\text{irrep}} \le \begin{cases} 10^{-10}, & \text{analytic}, \\ 0.02, & \text{sampled}. \end{cases} \tag{122}$$

For removable class defects, the analytic backend must reach matching collar classes within 20 repair activations for all six schedules; the sampled backend must do so within 24 repair activations in at least 90% of runs. The sampled nonabelian schedule bound is

$$D_{\text{sched}} \le 0.10. \tag{123}$$

6. **Harness gate.** Every negative control listed below must fail at least one of the gates above. If the negative controls do not fail, the validation harness is too weak to trust.

The scope labels should be reported explicitly. Passing the kinematic, record, and $\mathbb{Z}_2$ synchronization gates yields an `abelian-pilot-pass`. Passing those plus the nonabelian gate yields a `nonabelian-upgrade-pass`. A full `phase1-architecture-pass` requires both of those together with a passed harness gate.

10.5 Mandatory negative controls

The first validation package should contain the following explicit negative controls.

1. `NC-1-no-refresh`: remove Refresh_I, Refresh_J, and Refresh_{IJ} from the candidate menu. Expected failure: `Z2-1-stale-record` does not converge and $C_r(5)$ drops.

2. `NC-2-wrong-orientation`: reverse one stored edge orientation in the S_3 holonomy routine without updating the inverse lookup. Expected failure: M_{class} and D_{sched} jump while the $\mathbb{Z}_2$ pilot may look healthy.

3. `NC-3-no-illegal-penalty`: set $\lambda_u = 0$ in the S_3 build. Expected failure: u_{bad} and N_{irrep} become nonzero.

4. `NC-4-no-write`: skip the record-write layer. Expected failure: $C_r(5)$ and Q_{readback} fail even on clean runs.

10.6 Failure diagnostics

Every failed gate must be accompanied by a minimal evidence packet containing the benchmark identifier, schedule identifier, cycle index, pre- and post-correction register snapshots, the full candidate-cost table, the chosen correction, and the tie-break rank. The default diagnostic codes are:

1. **F1 gauge drift.** Signature: $g(t) > 0$ after L_{verify}. Likely causes: a correction move that does not preserve the Gauss sector, a wrong stored-edge ordering, or a verify layer that is reading the pre-repair state.
2. **F2 code leakage.** Signature: $u_{\text{bad}}(t) > 0$. Likely causes: an incomplete S_3 decode table, a missing invalid-word penalty, or a record class encoder that writes the illegal word 11.
3. **F3 record volatility.** Signature: low $C_r(5)$ with acceptable $g(t)$. Likely causes: the write layer is not pure overwrite, the record circuit reaches beyond the declared local support, or the readout channel is feeding back into nonshared plaquettes.
4. **F4 non-monotone repair.** Signature: $G_\Phi \leq 0$ or accepted moves that increase Φ. Likely causes: candidate costs evaluated on stale pre-repair data, a tie-break order that masks a lower-cost move, or a correction menu that is too small for the declared syndrome family.
5. **F5 schedule split.** Signature: small $g(t)$ and reasonable records but large D_{sched}. Likely causes: missing triple-overlap information, nonconfluent local tie-break rules, or an overlap observable family that is larger than the repair menu can support.
6. **F6 nonabelian label failure.** Signature: the $\mathbb{Z}_2$ pilot passes but M_{class} or N_{irrep} fails. Likely causes: an orientation mistake in face or collar holonomies, an incorrect multiplication table, or a bad character-projector implementation.
7. **F7 false obstruction or missed obstruction.** Signature: `Z2-4-frustrated-cycle` either converges spuriously to $\Phi = 0$ or never reaches a stable plateau. Likely causes: the residual-label logic is not implemented, or the compare layer is failing to distinguish removable mismatch from a persistent external fault source.

10.7 Reading rule for the validation package

Even a full reference-architecture pass would not identify a unique microscopic UV completion or prove the final repair law. It would only show that the fixed-cutoff architecture claimed here has been instantiated concretely enough to measure and debug. Conversely, failure is informative. A failed gate tells the implementation team which layer is not doing the job the paper assigns to it, and the required logs above tell them where to start looking. Literal microscopic uniqueness is not the target invariant here; the OPH theorem surface fixes only the physical branch modulo gauge or implementation hiding and inert ancillary refinement.

The analytic backend passes all named $\mathbb{Z}_2$ and S_3 fixed-cutoff gates across all six schedules. The pass is an existence-program benchmark for the regulated architecture. It does not settle uniqueness of the underlying microscopic completion and it does not replace packet-level quotient closure or long-run noisy approximate consensus for arbitrary exported packet nets.

11 Dependence on Existing OPH Results and Scope Boundaries

This note is intentionally one-sided in its logical role. It imports OPH structure that exists elsewhere in the program, adds one concrete fixed-cutoff reference architecture on top of that structure, and keeps the companion-paper theorem surfaces explicit rather than implicit. The intended claim boundary is therefore

$$\text{imported OPH framework} \longrightarrow \text{reference architecture} \longrightarrow \text{companion-paper closure surfaces.} \tag{124}$$

11.1 Imported OPH structure used here without re-derivation

The following ingredients are reused as background OPH structure. Their claim tier is inherited from the papers that establish them.

1. The screen-net viewpoint on a horizon screen S^2, together with patch algebras and the language of overlap consistency.
2. The collar and edge-center completion package: shared collar algebras, center/sector projectors Π_α, and the idea that overlap-accessible data live in a boundary-visible sector decomposition.
3. The recoverable generalized entropy and Markov/recovery vocabulary in which conditional mutual information, collar recovery, and edge data are meaningful organizing notions.
4. The local MaxEnt plus refinement-stability branch as the OPH dynamical background against which regulated architectures are later judged.
5. The support-visible BW scaling theorem together with the imported geometric-modular, null-modular, and local-Einstein closure surfaces used elsewhere in the corpus.
6. The OPH observer-access tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$, understood as an existing access model for a record-bearing internal subsystem.

11.2 Architecture-level content

Relative to that imported framework, the new content is at the architecture level rather than at the full completion level.

1. It selects one finite gauge-register or quantum-link style reference architecture for a cellulated screen, explicitly as a working model rather than as the unique OPH UV completion.
2. It gives an explicit fixed-cutoff dictionary from local registers to patch observables, overlap observables, collar-sector readouts, record variables, observer interfaces, and mismatch syndromes.
3. It specifies one fixed-cutoff repair interface

   $$\mathcal{R}_{PQ} = \mathcal{W}_{PQ} \circ \mathcal{C}_{PQ} \circ \mathcal{M}_{PQ} \circ \mathcal{P}_{PQ}, \tag{125}$$

   together with a finite local correction-menu template and an asynchronous update loop.
4. It proposes a concrete simulator ladder: an octahedral $\mathbb{Z}_2$ pilot first and an S_3 upgrade second.

5. It states a validation package that can fail, built from gauge-violation rate, overlap mismatch, record persistence, and schedule-divergence diagnostics.

These are fixed-cutoff regulator claims, simulator-interface claims, and theorem packages at the architecture level. They are not refinement-limit theorems, gravity theorems, or quantitative Standard Model closures.

11.3 Boundary markers between imported structure and new architecture content

Three boundaries matter for reading the paper correctly.

1. The existence of collar-sector observables is imported OPH regulator language; the new claim here is only that the chosen finite-register architecture can be organized so that a declared family of fixed-cutoff observables plays that interface role.

2. The tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$ is imported OPH observer language; the new content here is the added update interface $\mathcal{U}_O$ and the operational fixed-cutoff criterion for when a persistent pattern counts as an observer-like subsystem in simulation.

3. The need for overlap repair or synchronization is imported from the broader OPH consistency program; the new content here is only the explicit microscopic syntax of syndromes, local costs, and correction menus on a finite screen.

This paper does not claim complete OPH microphysics. It records one concrete regulated fixed-cutoff construction that carries the mapped fixed-cutoff theorem packages directly, while separating microscopic uniqueness and packet-level refinement questions from the fixed-cutoff construction.

11.4 Scope boundaries

The scope boundaries are:

1. **Microphysics uniqueness.** Nothing below claims that the octahedral $\mathbb{Z}_2$ pilot, the S_3 upgrade, or the thermalized edge-law branch are the unique OPH microphysics. They are explicit regulated fixed-cutoff constructions on which the observer-side theorem packages are carried and the structural branch boundaries are stated explicitly.

2. **Exported-packet repair and approximate stability.** The local repair kernel, the regulator embedding, the edge-law package, and the local measurement/backup packages are carried here as fixed-cutoff theorem packages. The consensus paper supplies exact finite repair/confluence, quotient normal forms, observable-level confluence on declared physical algebras, refinement-limit normal-form and holonomy classes under separated cofinal hypotheses, and controlled coarse-graining defects. The broader task is long-run noisy approximate consensus for arbitrary exported packet-net families.

The imported OPH structure stays imported, the finite reference architecture stays finite, the regulator-side patch-net embedding theorem, the finite-group edge-law theorem, and the three observer-side theorem packages are closed at fixed cutoff. Packet-level quotient closure beyond the declared exported domains, compact-group/Peter–Weyl lift bookkeeping for the D10 branch, and long-run noisy approximate consensus for arbitrary exported packet nets are separate downstream questions.

12 Fixed-Cutoff Theorem Packages

This paper proves five fixed-cutoff theorem packages: regulated patch-net embedding, edge heat-kernel / Casimir, measurement/Born-rule, Bell/CHSH, and observer checkpoint/restoration.

12.1 Regulated patch-net embedding theorem

Imported inputs. From the imported OPH background we use only the screen/patch/collar language, the existence of patch algebras and overlap-accessible collar sectors, and the observer-access tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$. The finite cellulation, the explicit patch-overlap graph, the boundary-fixed patch and collar algebras, the record registers, the packet readouts, and the local repair instruments are the regulated content established here.

Regulator-side object. On the fixed cellulation Γ and fixed patch cover $\mathcal{P} = \{P_I\}_{I\in\mathcal{V}}$, define the regulated OPH object

$$\mathfrak{R}_\Gamma^{\mathrm{OPH}} = \begin{pmatrix} \Gamma,\ \mathcal{G}_{\mathrm{patch}},\ \mathcal{H}_{\mathrm{phys}},\ \{\mathcal{A}_I\}_I, \\ \{\mathcal{A}_{IJ}\}_{\{I,J\}\in\mathcal{E}},\ \{\iota_{IJ\to I}, \iota_{IJ\to J}\}_{\{I,J\}\in\mathcal{E}}, \\ \{\mathcal{O}_{IJ}^{\mathrm{sh}}\}_{\{I,J\}\in\mathcal{E}},\ \{\mathfrak{I}_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \\ \{\mathcal{R}_{IJ}\}_{\{I,J\}\in\mathcal{E}},\ \{R_I\}_I \end{pmatrix}, \tag{126}$$

where $\mathcal{A}_I = \mathcal{B}(\widetilde{\mathcal{H}}_{P_I})^{G_{\partial P_I}}$ is the boundary-fixed patch algebra, $\mathcal{A}_{IJ} = \mathcal{B}(\widetilde{\mathcal{H}}_{B_{IJ}})^{G_{\partial B_{IJ}}}$ is the boundary-fixed overlap algebra, $\mathfrak{I}_{IJ}$ is the typed overlap object of the previous section, and $\mathcal{R}_{IJ}$ is the corresponding one-step local repair instrument on N_{IJ}.

Embedded finite patch-net layer. Define the finite patch-net layer carried by that regulator to be

$$\mathfrak{F}_\Gamma^{\mathrm{PN}} = \begin{pmatrix} \mathcal{G}_{\mathrm{patch}},\ \{\mathsf{X}_{IJ}\}_{\{I,J\}\in\mathcal{E}},\ \{\mathbf{q}_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \\ \{S_{IJ}\}_{\{I,J\}\in\mathcal{E}},\ \{L_{IJ}\}_{\{I,J\}\in\mathcal{E}},\ \{\Xi_{IJ}\}_{\{I,J\}\in\mathcal{E}}, \\ \{\mathcal{C}_{IJ}^{\min}\}_{\{I,J\}\in\mathcal{E}} \end{pmatrix}, \tag{127}$$

where

$$\mathbf{q}_{IJ}(\rho) = \big(\mathrm{read}_{I\to J}(\rho), \mathrm{read}_{J\to I}(\rho)\big) \tag{128}$$

is the joint overlap packet readout. Thus the finite patch-net data consist of explicit packet alphabets, explicit readout maps, explicit mismatch syndromes, explicit local scores, and an explicit finite correction menu on every overlap edge.

Consensus-format export. For each patch P_I, let

$$\Pi_I^{\mathrm{pkt}}(\rho) = \big(x_{I\to J}^{(IJ)}(\rho)\big)_{J:\{I,J\}\in\mathcal{E}} \in \prod_{J:\{I,J\}\in\mathcal{E}} \mathsf{X}_{IJ} \tag{129}$$

be the full outgoing packet tuple exported by that patch on all incident overlaps, and define the finite patch-state space

$$S_I^{\mathrm{reg}} := \operatorname{im} \Pi_I^{\mathrm{pkt}}. \tag{130}$$

For each overlap edge $e = \{I, J\}$, define the interface alphabet

$$I_e^{\mathrm{reg}} := \mathsf{X}_{IJ}, \tag{131}$$

and let

$$\pi^{\mathrm{reg}}_{I,e} : S^{\mathrm{reg}}_I \to I^{\mathrm{reg}}_e, \qquad \pi^{\mathrm{reg}}_{J,e} : S^{\mathrm{reg}}_J \to I^{\mathrm{reg}}_e \tag{132}$$

be the coordinate projections onto the IJ packet slots. Thus the abstract patch-net datum

$$(\mathcal{G}_{\mathrm{patch}}, \{S_I\}_I, \{I_e\}_e, \{\pi_{I,e}\}_{I,e})$$

used later in the consensus paper has a literal regulator-side export here. The separate consensus question concerns quotient dynamics: when a declared observable family passes packet closure, the exact regulator repair instrument pushes forward to an autonomous packet-level update rule; when it fails, the hidden-state dependence is logged and no such quotient dynamics is claimed.

Embedding theorem. The finite patch-net layer $\mathfrak{F}^{\mathrm{PN}}_{\Gamma}$ is explicitly realized inside the fixed-cutoff OPH regulator $\mathfrak{R}^{\mathrm{OPH}}_{\Gamma}$. Concretely:

1. every vertex of the finite patch net is a genuine patch P_I of the cellulated screen and every edge is a genuine overlap collar B_{IJ};
2. every packet field is the output of a declared finite readout on regulator observables in $\mathcal{A}_{IJ}$ and on the declared record layer;
3. the consensus-format state spaces S^{reg}_I, interface alphabets $I^{\mathrm{reg}}_{\{I,J\}}$, and projection maps $\pi^{\mathrm{reg}}_{I,\{I,J\}}$ are exported from those same readouts;
4. every mismatch syndrome, local score, and allowed correction branch is defined from those same readouts and from the same local repair instrument $\mathcal{R}_{IJ}$.

So the finite patch-net objects used later in the consensus lane are not abstract placeholders. They are actual regulator-side data on one explicit finite screen model.

Proof. The patch cover and overlap graph are finite because the underlying cellulation is finite. The patch algebras $\mathcal{A}_I$ and overlap algebras $\mathcal{A}_{IJ}$ are explicit finite-dimensional boundary-fixed algebras, so the regulator presentation required for a fixed-cutoff OPH patch net is in hand. The collar-sector projectors $\Pi^{(IJ)}_{\alpha}$ and the rest of the declared shared observable family $\mathcal{O}^{\mathrm{sh}}_{IJ}$ live in those overlap algebras, hence the packet alphabets X_{IJ} are realized by actual finite readouts rather than by external labels. The combined outgoing packet map Π^{pkt}_I therefore has finite image, so the exported patch-state spaces S^{reg}_I and interface projections $\pi^{\mathrm{reg}}_{I,\{I,J\}}$ are finite regulator-side objects as well. The candidate menu $\mathcal{C}^{\mathrm{min}}_{IJ}$ consists of concrete local link, face, or record-refresh branches supported on N_{IJ}, and the staged map

$$\mathcal{R}_{IJ} = \mathcal{W}_{IJ} \circ \mathcal{C}_{IJ} \circ \mathcal{M}_{IJ} \circ \mathcal{P}_{IJ} \tag{133}$$

is therefore a concrete local repair instrument on the same regulator. The mismatch syndrome S_{IJ}, the lexicographic score L_{IJ}, and the diagnostic cost Ξ_{IJ} are explicit functions of the packet values exported by $\mathbf{q}_{IJ}$. Nothing in this construction invokes refinement-limit lift, global fixed points, or refinement-stable consensus. Therefore the finite patch-net layer is embedded into the OPH regulator at fixed cutoff.

Scope boundary. What is proved here is an *object-level and local-interface embedding theorem.* It says that the finite patch-net variables, comparison rules, and local repair syntax are all realized by the regulated screen microphysics. Autonomous quotient dynamics for the packet layer requires the packet-closure test introduced in the preceding section: identical input packets must lead to identical reread packets after every accepted branch. When that test passes for a declared observable family, the regulator pushes forward to the exact finite packet-level update rule on that family. If the test fails, the embedding proved here stands. Packet quotient closure is therefore a branch-local export condition rather than part of the object-level embedding theorem.

Conclusion for this package. The regulated patch-net embedding is explicit at fixed cutoff: patch vertices, patch-state spaces, interface alphabets, projection maps, syndrome components, and candidate menus are fixed by the regulator. The consensus paper proves repair completeness, quotient compatibility, and the classical full-support Petz support-gap clause on a rooted-tree packet-net export. Gauge-invariant observables then factor through the quotient normal form on that verified branch. For a general regulator export, the downstream items are packet-level quotient closure when available, repair completeness for that declared finite patch-net branch, the quotient-level compatibility package that yields the local diamond, residual-obstruction classification, and refinement-limit normal-form/holonomy transport under the required naturality and separation hypotheses. On a verified packet-closed branch, the consensus theorem package yields Lyapunov descent, derived termination, local confluence, schedule robustness, and normal-form existence.

12.2 Edge heat-kernel / Casimir theorem

Inputs used here. From the earlier fixed-cutoff microphysics we use only the same declared overlap interface as in the reference-model section: the fixed patch cover and overlap collars, the cut register h_{IJ}, the sector projectors $\{\Pi_R^{(IJ)}\}_R$, the exact edge-state and local-sampler modes, and the same gauge-preserving local move sets on N_{IJ} from which the repair and thermal branches are built. On the compact-group truncation branch we also use the declared edge operator

$$\Delta_{IJ}^{\mathrm{QL}} = \sum_{R\in\mathcal{R}_{\max}} C_2(R)\,\Pi_R^{(IJ)}.$$

No refinement-limit lift, no global consensus theorem, and no separate D10 calibration package are used here. The theorem-local kernel condition is that the thermal proposal kernel induced by the declared local move set obeys the reversible symmetry $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$ and is irreducible/aperiodic on the connected finite sector component under study.

Thermalized edge-law branch. Use the same regulated collars, the same overlap sector projectors $\{\Pi_\alpha^{(IJ)}\}_\alpha$, and the same correction menus as above, but replace the deterministic "commit the lexicographic minimizer" rule on one active overlap by a reversible thermal branch. For sectors α, β on one overlap, let $q_{\alpha\beta}$ be the microphysical proposal kernel induced by the local generator menu, with the symmetry property

$$d_\alpha\, q_{\alpha\beta} = d_\beta\, q_{\beta\alpha}, \tag{134}$$

and let the acceptance rule be

$$A_{\alpha\to\beta} = \min\Big(1,\, \exp\big[-\beta\big(C_2(\beta) - C_2(\alpha)\big)\big]\Big), \tag{135}$$

where $C_2(\alpha)$ is the quadratic Casimir of the sector label carried by $\Pi_\alpha^{(IJ)}$ and $\beta > 0$ is the overlap inverse temperature parameter of the branch. The one-step overlap kernel is then

$$K_{\alpha\beta} = \begin{cases} q_{\alpha\beta} A_{\alpha\to\beta}, & \alpha \neq \beta, \\ 1 - \sum_{\gamma\neq\alpha} q_{\alpha\gamma} A_{\alpha\to\gamma}, & \alpha = \beta. \end{cases} \tag{136}$$

Finite-group Casimir theorem. The stationary sector law of the thermalized overlap branch is

$$\pi_\beta(\alpha) = \frac{d_\alpha e^{-\beta C_2(\alpha)}}{Z_\beta}, \qquad Z_\beta = \sum_\gamma d_\gamma e^{-\beta C_2(\gamma)}. \tag{137}$$

Thus the explicit regulated microphysics carries the edge-sector Casimir law rather than only logging candidate sector frequencies.

Proof. By construction,

$$\pi_\beta(\alpha) K_{\alpha\beta} = \frac{d_\alpha e^{-\beta C_2(\alpha)}}{Z_\beta} q_{\alpha\beta} \min\Big(1,\, e^{-\beta(C_2(\beta)-C_2(\alpha))}\Big) \tag{138}$$

$$= \frac{d_\beta e^{-\beta C_2(\beta)}}{Z_\beta} q_{\beta\alpha} \min\Big(1,\, e^{-\beta(C_2(\alpha)-C_2(\beta))}\Big) = \pi_\beta(\beta) K_{\beta\alpha}, \tag{139}$$

using the proposal symmetry $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$. So detailed balance holds, hence π_β is invariant. Because the local generator menu connects the finite pilot sectors and the kernel has nonzero self-transition, the chain is irreducible and aperiodic on each connected sector component, so this invariant law is the unique stationary distribution there.

Heat-kernel corollary. Along any refinement ladder whose finite-group sector data converge to a compact-group Peter–Weyl decomposition, the same stationary weights converge cylinderwise to the standard heat-kernel / Casimir form

$$w_\beta(R) \propto d_R e^{-\beta C_2(R)} \tag{140}$$

on irreducible representation labels R. Thus the explicit screen microphysics carries the SM/GR paper's edge-law branch. Microscopic uniqueness and model-family universality remain separate questions.

Fixed-cutoff closure. The previous reference-model section supplies the concrete same-overlap realization needed for audit here: a declared cut register on each overlap, explicit sector projectors, an exact edge-state mode, a local sampler mode, and a compact-group truncation branch with $\Delta_{IJ}^{\mathrm{QL}} = \sum_R C_2(R)\Pi_R^{(IJ)}$. So the theorem above is not a detached Markov-chain exercise. It is the explicit fixed-cutoff edge law of the same regulated overlap microphysics used for synchronization and repair.

Merge boundary. What is closed here is exactly the fixed-cutoff edge-law package on the declared overlap interface: the cut register, the sector projectors, the reversible thermal branch, and the finite-group stationary Casimir law, together with the Peter–Weyl heat-kernel corollary when the stated refinement ladder is supplied. The later compact D10 surfaces import this package through that declared handoff. They do not need a new leaf-level Casimir derivation; the compact-group / Peter–Weyl lift belongs to the companion D10 and compact-branch bookkeeping, while large-N_{edge} and critical-string claims belong to the continuation lane.

Selector benchmark. This same handoff supplies the equilibrium benchmark for the D10 calibration lane. Writing

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)},$$

the exact self-similar balance condition

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle}$$

forces $x^2 - x - 1 = 0$ and hence $x = \varphi = (1 + \sqrt{5})/2$. Equivalently $A_\varphi(x) = x - 1 - \frac{1}{x}$ vanishes at the exact equilibrium point. The exact numerical value of P is fixed on the calibration surface. The point of the benchmark is that the proximity to φ has a structural explanation. Exact balance is too symmetric to support durable records, structure, and dynamics. The microphysics handoff identifies the balance point, and the D10 solve fixes the realized detuning.

Conclusion for this package. This fixes the edge heat-kernel / Casimir package at fixed cutoff. Microscopic uniqueness and model-family universality are separate questions. The leaf-level Casimir derivation is supplied on the declared fixed-cutoff surface.

12.3 Measurement / Born-rule theorem stack

Inputs used here. From the earlier fixed-cutoff microphysics we use only the same declared observer-facing local readout surface and center structure: the pointer projectors on record registers, the overlap-sector projectors on the same collar interfaces, and the completed compare, write, and verify slice on which those observables are read. No refinement-limit lift, no extra observer-metaphysics, and no separate consensus theorem are used here. The appendix ledger sharpens this package with Markov/recoverability and Gleason-side comparisons, but the fixed-cutoff closure stated in this subsection is carried on the main microphysics surface itself.

Record algebra. After each completed write/verify phase, let

$$\mathcal{Z}_{\text{rec}}(t) = \text{Alg}\Big(\{P_{m_I^{(J)}(t)}\}_{I \neq J}, \{P_{r_I^{\text{bulk}}(t)}\}_I, \{\Pi_\alpha^{(IJ)}(t)\}_{I<J,\alpha}\Big) \tag{141}$$

be the finite record algebra generated by the declared pointer projectors and the overlap-sector projectors exposed to the observer-accessible collar interfaces.

Centrality proposition. Within one completed compare/write/verify slice, $\mathcal{Z}_{\text{rec}}(t)$ is a commutative subalgebra and commutes with every declared readout observable used by the measurement instrument.

Proof. Every generator of $\mathcal{Z}_{\text{rec}}(t)$ is, by construction, a projector on a finite classical register value or a direct sum of such values. The compare layer reads only those same declared register/projector outcomes, and the verify layer only re-reads them. Hence the entire instrument relevant to fixed-cutoff measurement factors through one finite classical algebra. So the record algebra is central for the operational measurement surface carried here.

Born-rule theorem. For every event E in the finite sigma algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$, the measurement probability is

$$\mathbb{P}_t(E) = \mathrm{Tr}(\rho_t P_E), \tag{142}$$

where $P_E \in \mathcal{Z}_{\mathrm{rec}}(t)$ is the projector of the event. If one conditions on E and then continues the run with the same subsequent update schedule, the conditioned state is

$$\rho_t|_E = \frac{P_E \rho_t P_E}{\mathrm{Tr}(\rho_t P_E)}, \tag{143}$$

so the operational state update is precisely Born/Lüders conditioning on the central record algebra.

Proof. The completed write/verify slice defines a finite projective instrument on the commuting projectors of $\mathcal{Z}_{\mathrm{rec}}(t)$. In finite dimension, the probability of an event projector is its Born trace, and conditioning on that same central projector produces the Lüders update written above. Because the record algebra is central for the declared readout surface, no extra hidden noncommutative ambiguity survives at the measurement interface.

Repeated-read corollary. If no accepted repair move between t and $t+1$ touches the support of P_E, then the next read of the same event has probability 1. If the support is touched, the cycle table records the precise place where the conditioning surface changed. The measurement package is defined with an auditable conditioning surface.

Merge boundary. What is closed here is exactly the fixed-cutoff operational measurement package: central record algebra, Born probabilities on its event projectors, Lüders conditioning on that same algebra, and repeated-read stability only under the explicit untouched-support hypothesis above. The later supplement may carry sharper Markov/recoverability and comparison-theorem bookkeeping, and the observer synthesis paper may import this package as theorem-bearing on the same operational surface, but neither move upgrades the claim into a broader resolution of measurement. There is therefore no hidden supplement- or observer-paper-merge blocker on this leaf; the boundary is only between this fixed-cutoff theorem stack and later interpretive or global-observer continuations.

Conclusion for this package. This closes the fixed-cutoff measurement/Born-rule package on the microphysics surface: the record algebra, Born trace, Lüders conditioning, and the limited repeated-read corollary are all fixed on the declared operational surface.

12.4 Bell / CHSH theorem stack

Inputs used here. From the fixed-cutoff theorem surfaces we use only the declared physical observable algebra on one observer-facing two-wing region, the central setting registers on the same compare slice, and the Born/Lüders package on the resulting commuting event algebra. The consensus paper supplies the quotient-level physical observable algebra and observable-level confluence on that same fixed-cutoff branch. The only downstream blocker named here is the later refinement-limit lift where the physical observable algebra itself changes under refinement.

Bell data on the fixed-cutoff surface. Fix two disjoint observer-facing wing regions L and R on one declared fixed-cutoff physical algebra $\mathcal{A}_{\text{phys}}(L \cup R)$, and let

$$\mathcal{A}_L, \mathcal{A}_R \subset \mathcal{A}_{\text{phys}}(L \cup R)$$

be commuting subalgebras,

$$[X_L, X_R] = 0 \qquad \forall\, X_L \in \mathcal{A}_L,\ X_R \in \mathcal{A}_R.$$

Let $x, y \in \{0, 1\}$ be classical setting values written on central setting registers, and for each setting choose binary projective readouts

$$\{P^{(L)}_{a|x}\}_{a=\pm 1} \subset \mathcal{A}_L, \qquad \{P^{(R)}_{b|y}\}_{b=\pm 1} \subset \mathcal{A}_R,$$

with

$$P^{(L)}_{a|x} P^{(L)}_{a'|x} = \delta_{aa'} P^{(L)}_{a|x}, \qquad \sum_{a=\pm 1} P^{(L)}_{a|x} = \mathbf{1},$$
$$P^{(R)}_{b|y} P^{(R)}_{b'|y} = \delta_{bb'} P^{(R)}_{b|y}, \qquad \sum_{b=\pm 1} P^{(R)}_{b|y} = \mathbf{1}.$$

Assume the source preparation is a physical state ρ_{LR} on $\mathcal{A}_{\text{phys}}(L \cup R)$. The source-state input is explicit here: this theorem does not derive Bell-pair preparation from the bare repair dynamics.

Bell-law theorem. For every fixed setting pair (x, y), the joint event projectors

$$\Pi_{ab|xy} := P^{(L)}_{a|x} P^{(R)}_{b|y}$$

belong to one commuting event algebra, and the outcome law on the compare slice is

$$p(a, b \mid x, y) = \operatorname{Tr}\!\big(\rho_{LR}\, \Pi_{ab|xy}\big) = \operatorname{Tr}\!\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big). \tag{144}$$

This law is positive, normalized, and no-signaling:

$$\sum_{b=\pm 1} p(a, b \mid x, y) = \operatorname{Tr}\!\big(\rho_{LR}\, P^{(L)}_{a|x}\big), \qquad \sum_{a=\pm 1} p(a, b \mid x, y) = \operatorname{Tr}\!\big(\rho_{LR}\, P^{(R)}_{b|y}\big), \tag{145}$$

so the left marginal is independent of y and the right marginal is independent of x.

Proof. Because the left and right projectors commute, $\Pi_{ab|xy}$ is a commuting projective family on the fixed setting slice. The Born theorem proved above on the same commuting event algebra therefore gives the joint law. Positivity is immediate, and normalization follows from

$$\sum_{a,b} \Pi_{ab|xy} = \left(\sum_a P^{(L)}_{a|x}\right) \left(\sum_b P^{(R)}_{b|y}\right) = \mathbf{1}.$$

Summing over one wing uses the same completeness relation and removes the remote projector, which proves the no-signaling identities.

CHSH/Tsirelson theorem. Define the binary observables

$$A_x := P^{(L)}_{+|x} - P^{(L)}_{-|x}, \qquad B_y := P^{(R)}_{+|y} - P^{(R)}_{-|y}.$$

Then

$$A_x^2 = \mathbf{1}, \qquad B_y^2 = \mathbf{1}, \qquad [A_x, B_y] = 0,$$

and the Bell correlators are

$$E(x,y) := \sum_{a,b=\pm 1} ab\, p(a,b \mid x,y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big). \tag{146}$$

For the CHSH operator

$$\mathcal{B} := A_0(B_0 + B_1) + A_1(B_0 - B_1), \tag{147}$$

one has

$$\mathcal{B}^2 = 4\,\mathbf{1} - [A_0, A_1][B_0, B_1]. \tag{148}$$

Hence

$$\|\mathcal{B}\| \le 2\sqrt{2}, \qquad \big|E(0,0) + E(0,1) + E(1,0) - E(1,1)\big| \le 2\sqrt{2}. \tag{149}$$

Proof. The correlator formula is the usual projector expansion,

$$A_x B_y = \sum_{a,b=\pm 1} ab\, P^{(L)}_{a|x} P^{(R)}_{b|y}.$$

Expanding $\mathcal{B}^2$ and using $A_x^2 = B_y^2 = \mathbf{1}$ together with $[A_x, B_y] = 0$ gives the identity above. Because $\|A_x\|, \|B_y\| \le 1$,

$$\|[A_0, A_1]\| \le 2, \qquad \|[B_0, B_1]\| \le 2,$$

so $\|\mathcal{B}^2\| \le 8$ and therefore $\|\mathcal{B}\| \le 2\sqrt{2}$. Taking the expectation value in ρ_{LR} gives the CHSH bound.

Two-qubit saturation corollary. Assume additionally that there exist commuting unital $*$-subalgebras

$$\mathcal{Q}_L \subset \mathcal{A}_L, \qquad \mathcal{Q}_R \subset \mathcal{A}_R,$$

with

$$\mathcal{Q}_L \cong M_2(\mathbb{C}), \qquad \mathcal{Q}_R \cong M_2(\mathbb{C}), \qquad C^*(\mathcal{Q}_L, \mathcal{Q}_R) \cong M_2(\mathbb{C}) \otimes M_2(\mathbb{C}).$$

Assume that, under this identification, the restriction of ρ_{LR} to $C^*(\mathcal{Q}_L, \mathcal{Q}_R)$ is

$$|\Phi^+\rangle\langle\Phi^+|, \qquad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

and that the fixed-cutoff observables restrict there to

$$A_0 = \sigma_z \otimes \mathbf{1}, \qquad A_1 = \sigma_x \otimes \mathbf{1},$$

$$B_0 = \mathbf{1} \otimes \frac{\sigma_z + \sigma_x}{\sqrt{2}}, \qquad B_1 = \mathbf{1} \otimes \frac{\sigma_z - \sigma_x}{\sqrt{2}}.$$

Then

$$E(0,0) = E(0,1) = E(1,0) = \frac{1}{\sqrt{2}}, \qquad E(1,1) = -\frac{1}{\sqrt{2}},$$

and hence

$$E(0,0) + E(0,1) + E(1,0) - E(1,1) = 2\sqrt{2}.$$

Proof. This is the standard two-qubit computation on the explicitly assumed $M_2(\mathbb{C}) \otimes M_2(\mathbb{C})$ subalgebra. It is an exact branch-level statement on that added two-qubit input; it does not derive tensor-factor structure or Bell-pair preparation from the bare OPH repair dynamics.

Merge boundary. What is closed here is exactly the fixed-cutoff Bell package on the declared two-wing surface: source-specified joint law, no-signaling on commuting separated observables, the CHSH correlator formula, and the Tsirelson bound. Exact $2\sqrt{2}$ saturation is fixed only on the explicit two-qubit branch above. What is not claimed here is a derivation of Bell-pair source preparation or tensor-factor structure from the bare repair dynamics, or a refinement-stable Bell package beyond the fixed-cutoff physical observable algebra.

Conclusion for this package. This closes the fixed-cutoff Bell/CHSH package for the source-specified two-wing law, separated-observable no-signaling, and the Tsirelson bound. Exact $2\sqrt{2}$ saturation is fixed only under the explicit two-qubit branch input above.

12.5 Observer continuation / backup theorem stack

Checkpoint data. For an observer patch P_O, define the checkpoint at cycle t to be

$$\mathrm{Chk}_O(t) = \big(\mathcal{Z}_{\mathrm{rec}}^{(O)}(t), \rho_O^{\mathrm{acc}}(t), \nu_{\geq t}, \mathfrak{I}_O^{\mathrm{ext}}(t)\big), \tag{150}$$

where $\mathcal{Z}_{\mathrm{rec}}^{(O)}(t)$ is the restriction of the record algebra to the observer-accessible record/pointer registers, $\rho_O^{\mathrm{acc}}(t)$ is the observer-patch state restricted to the algebra generated by those registers and the declared observer-facing collar observables, $\nu_{\geq t}$ is the future update schedule, and $\mathfrak{I}_O^{\mathrm{ext}}(t)$ is the tuple of externally exposed overlap interfaces incident on P_O.

Restoration map. A restoration map at accuracy ε is any channel

$$\mathcal{B}_\varepsilon : \mathrm{Chk}_O(t) \longmapsto \widetilde{\rho}_O(t) \tag{151}$$

such that

$$\left\|\widetilde{\rho}_O^{\mathrm{acc}}(t) - \rho_O^{\mathrm{acc}}(t)\right\|_1 \leq \varepsilon \tag{152}$$

and the restored run begins with the same future schedule $\nu_{\geq t}$ and the same externally visible interface data $\mathfrak{I}_O^{\mathrm{ext}}(t)$.

Continuation theorem. If $\varepsilon = 0$, then every future observer-accessible event has exactly the same probability law on the restored run as on the original run. More generally, for any $\varepsilon \geq 0$, the total-variation distance between the two future history laws on the observer-accessible event algebra is at most ε.

Proof. All future observer-accessible probabilities are generated by repeatedly applying completely positive trace-preserving update maps and projective readouts to the accessible algebra of P_O together with the fixed external overlap interface data. Exact restoration therefore gives the same initial state on the same accessible algebra and the same future channel sequence, hence identical future laws. For approximate restoration, contractivity of trace distance under completely positive trace-preserving maps gives

$$\left\|\mathcal{E}(\widetilde{\rho}_O^{\mathrm{acc}}) - \mathcal{E}(\rho_O^{\mathrm{acc}})\right\|_1 \leq \left\|\widetilde{\rho}_O^{\mathrm{acc}} - \rho_O^{\mathrm{acc}}\right\|_1 \leq \varepsilon \tag{153}$$

for every later channel $\mathcal{E}$, and therefore the induced difference of all later event probabilities is bounded by ε.

Identity criterion. Two checkpoints represent the same observer in this paper iff they induce the same future law on the observer-accessible event algebra. Exact sameness means equality of that law; ε-approximate sameness means total-variation distance at most ε. Thus the observer-identity criterion is operational and testable rather than rhetorical.

Backup corollary. Any exact copy of $\mathrm{Chk}_O(t)$ is a valid backup. Any approximate copy satisfying the same restoration bound is an ε-backup with explicit quantitative degradation.

Conclusion for this package. This closes the fixed-cutoff observer checkpoint/restoration package: the checkpoint, restoration, error, and identity statements above are fixed on the microphysics surface.

12.6 Fixed-Cutoff Boundaries and Downstream Questions

The downstream questions are:

1. The regulated patch-net embedding package is closed here at the object and local-interface level. Packet-level quotient closure on a declared observable family and long-run noisy approximate consensus on that exported packet net are separate tasks.
2. The edge heat-kernel / Casimir package has an explicit thermalized finite-group theorem here. The compact-group / Peter–Weyl lift is handled on the companion compact and D10 surfaces, and large-N_{edge} worldsheet statements use their own continuation inputs.
3. The fixed-cutoff measurement/Born-rule package is closed here by the central record algebra, Born trace, and conditioning theorem stack.
4. The fixed-cutoff Bell/CHSH package is closed here on the declared two-wing surface by the source-specified joint law, no-signaling theorem, and CHSH/Tsirelson theorem stack.
5. The fixed-cutoff observer checkpoint/restoration package is closed here by the checkpoint, restoration, error, and identity theorem stack.
6. What sits after these fixed-cutoff results is packet-level quotient closure beyond declared exported domains, model-family universality, and long-run noisy approximate consensus for arbitrary exported packet nets.

13 Conclusion

The regulated screen architecture carries five fixed-cutoff theorem packages:

1. the regulated patch-net embedding theorem, with the packet-closure and long-run noisy-consensus boundary made explicit,
2. the edge-sector thermal/Casimir theorem, with the compact-group / Peter–Weyl handoff made explicit,
3. the central-record/Born-rule measurement package,
4. the fixed-cutoff Bell/CHSH package on the declared two-wing surface, and
5. the checkpoint/restoration/backup observer package.

These results do *not* identify the unique OPH UV completion or prove long-run noisy approximate consensus for arbitrary exported packet nets. The regulated patch-net embedding theorem is closed on its fixed-cutoff microphysics surface; downstream work concerns packet-level quotient closure beyond declared exported domains and model-family universality.

A Formal Measurement / Markov / Born-Lüders Ledger

This appendix carries the formal measurement-theory block for the microphysics paper. It sharpens the fixed-cutoff record interface used in the main theorem packages above. The theorem-bearing leaf-level closure is the main-text fixed-cutoff central-record / Born-Lüders package; this appendix is the supporting ledger and comparison surface, not a replacement theorem surface.

A.1 Records as Central Variables

Definition. On one completed observer-facing slice in patch P, an *exact record presentation* is a family of orthogonal projectors $\{\widehat{\Pi}_r\} \subset Z(\mathcal{A}(P))$ whose generated algebra

$$\mathcal{R} := \mathrm{vN}(\{\widehat{\Pi}_r\})$$

is finite and commutative. A *practical record presentation* is a family of projectors $\{\Pi_r\} \subset \mathcal{A}(P)$ on the same declared slots such that

$$\delta_{\mathrm{rec}} := \max_r \|\Pi_r - \widehat{\Pi}_r\|$$

is finite. The exact central presentation carries the theorem-bearing fixed-cutoff measurement surface; the practical presentation records controlled readout error above it. The declared record surface must also satisfy:

1. **Stability:** the declared readout projectors on that slice are stable under the effective fixed-cutoff dynamics.
2. **Shareability:** each $\widehat{\Pi}_r$ lies in the center of the relevant overlap algebra.

Proposition 3.0 (Quantitative record stability). For a practical record presentation on the same declared slots,

$$\|[\Pi_r, \Pi_s]\| \le 4\,\delta_{\mathrm{rec}}$$

for all r, s. If $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$, then each declared elementary record-event probability obeys

$$\left|\mathrm{Tr}(\widetilde{\rho}\,\Pi_r) - \mathrm{Tr}(\rho\,\widehat{\Pi}_r)\right| \le \varepsilon + \delta_{\mathrm{rec}}.$$

Proof sketch. Because $[\widehat{\Pi}_r, \widehat{\Pi}_s] = 0$, one writes

$$[\Pi_r, \Pi_s] = [\Pi_r - \widehat{\Pi}_r, \Pi_s] + [\widehat{\Pi}_r, \Pi_s - \widehat{\Pi}_s],$$

and bounds each term by twice the corresponding operator-norm error. The probability estimate is the same two-term decomposition used on the main-text record surface:

$$\mathrm{Tr}(\widetilde{\rho}\,\Pi_r) - \mathrm{Tr}(\rho\,\widehat{\Pi}_r) = \mathrm{Tr}((\widetilde{\rho} - \rho)\Pi_r) + \mathrm{Tr}(\rho(\Pi_r - \widehat{\Pi}_r)),$$

so trace-norm control contributes ε and projector error contributes δ_{rec}.

A.2 Markov Structure and Recoverability

Theorem 3.1 (Quantum Markov chain structure on a fixed EC collar). If $I(A:C\mid B)=0$ for a tripartite state ρ_{ABC}, then

$$\mathcal{H}_B \cong \bigoplus_j \left(\mathcal{H}_{b_L^{(j)}} \otimes \mathcal{H}_{b_R^{(j)}}\right)$$

and the state decomposes as

$$\rho_{ABC} = \bigoplus_j q_j\, \rho^{(j)}_{Ab_L^{(j)}} \otimes \rho^{(j)}_{b_R^{(j)}C}.$$

The label j is a classical variable living in the center of $\mathcal{A}(B)$.

On the fixed finite-dimensional collar Hilbert space supplied by EC, let

$$\mathfrak{M}_{A:B:C} := \{\sigma_{ABC} : I(A:C\mid B)_\sigma = 0\}$$

and define the exact-Markov distance modulus

$$\delta^{\mathrm{M}}_{A:B:C}(\varepsilon) := \sup\left\{\inf_{\sigma\in\mathfrak{M}_{A:B:C}} \|\rho-\sigma\|_1 : I(A:C\mid B)_\rho \le \varepsilon\right\}.$$

Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:C}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Approximate collars therefore converge to the exact HJPW normal form on one fixed finite-dimensional collar model or on a compatible pulled-back family. This appendix does not claim a universal dimension-free one-shot trace-norm bound from small $I(A:C\mid B)$ directly to an exact Markov state.

Theorem 3.2 (Fawzi–Renner recovery). If

$$I(A:C\mid B)_\rho \le \varepsilon,$$

then there exists a CPTP map $\mathcal{R}:\mathcal{A}(B)\to\mathcal{A}(BC)$ such that the recovered comparison state

$$\rho^{\mathrm{rec}}_{ABC} := (\mathrm{id}_A\otimes\mathcal{R})(\rho_{AB})$$

obeys

$$\|\rho_{ABC}-\rho^{\mathrm{rec}}_{ABC}\|_1 \le 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

If ε is quoted in bits rather than nats, this bound is equivalently written as $2\sqrt{\ln 2\,\varepsilon}$.

The recovered state is a constructive comparison state and need not be exact Markov. To connect approximate recoverability to the exact splice or additivity formulas used later, one combines the recovery theorem with the fixed-collar modulus above. Concretely, if $\sigma_\varepsilon \in \mathfrak{M}_{A:B:C}$ is chosen so that

$$\|\rho-\sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:C}(\varepsilon),$$

then every bounded observable supported on the left side of the collar differs from its exact Markov splice value by at most

$$\|X\|_\infty\, \delta^{\mathrm{M}}_{A:B:C}(\varepsilon),$$

and, under uniform faithfulness with lower spectral bound $\lambda_* > 0$, the corresponding modular-additivity defect differs from the exact Markov value by at most

$$4\lambda_*^{-1}\,\delta^{\mathrm{M}}_{A:B:C}(\varepsilon).$$

So the microphysics paper keeps two distinct errors visible:

$$r_{\mathrm{FR}}(\varepsilon) = O(\varepsilon^{1/2}), \qquad \delta^{\mathrm{M}}_{A:B:C}(\varepsilon) \to 0.$$

A.3 Definiteness, Born, and Conditioning

Theorem 3.3 (Collapse from EC structure). Under EC, the post-measurement physical state is exactly a convex mixture over sector labels α:

$$\rho = \bigoplus_{\alpha} p_\alpha\, \rho^{(\alpha)}.$$

For any observer whose accessible observables lie in either side algebra:

1. there are no interference observables between different α-blocks; and
2. each observer-instance has a definite α in the classical sense.

Cross-block operators $|i\rangle\langle j|$ for $i \in \alpha$, $j \in \alpha'$, and $\alpha \neq \alpha'$ are not gauge-invariant and therefore do not belong to the physical algebra.

Theorem 3.4 (Gleason). For $\dim \mathcal{H} \geq 3$, any measure μ on projectors satisfying:

1. noncontextuality: $\mu(P)$ depends only on P;
2. additivity: $\mu(P \vee Q) = \mu(P) + \mu(Q)$ for $P \perp Q$; and
3. normalization: $\mu(\mathbf{1}) = 1$,

must have the form

$$\mu(P) = \mathrm{Tr}(\rho P)$$

for some density operator ρ.

Corollary 3.5 (Born uniqueness on the fixed-cutoff record surface). Within the fixed-cutoff type-I regulator presentation used here, Born's rule is the unique overlap-consistent probability assignment on the record algebra.

Proposition 3.6 (Luders update). For record projectors $\{\Pi_r\}$ and state ω,

$$p_r = \omega(\Pi_r), \qquad \omega_r(X) = \frac{\omega(\Pi_r X \Pi_r)}{\omega(\Pi_r)}.$$

Then for all X commuting with the record algebra $\mathcal{R}$,

$$\omega(X) = \sum_r p_r\, \omega_r(X).$$

So collapse on the observer-facing fixed-cutoff surface is simply classical conditioning on the commuting record algebra.

A.4 Two-Wing Bell / CHSH Package

Definition. On one declared fixed-cutoff physical observable algebra $\mathcal{A}_{\rm phys}(L \cup R)$, a *two-wing Bell presentation* consists of:

1. commuting left/right wing subalgebras
$$\mathcal{A}_L, \mathcal{A}_R \subset \mathcal{A}_{\rm phys}(L \cup R), \qquad [X_L, X_R] = 0 \quad \forall\, X_L \in \mathcal{A}_L,\ X_R \in \mathcal{A}_R;$$
2. central classical setting values $x, y \in \{0, 1\}$ on the declared record slots;
3. binary projective readouts
$$\{P^{(L)}_{a|x}\}_{a=\pm1} \subset \mathcal{A}_L, \qquad \{P^{(R)}_{b|y}\}_{b=\pm1} \subset \mathcal{A}_R,$$
with orthogonality and completeness on each wing; and
4. a physical source state ρ_{LR} on $\mathcal{A}_{\rm phys}(L \cup R)$.

The source-state input is explicit. This appendix does not derive Bell-pair preparation from bare repair dynamics.

Theorem 3.7 (Fixed-cutoff Bell law and no-signaling). For a two-wing Bell presentation, define
$$\Pi_{ab|xy} := P^{(L)}_{a|x} P^{(R)}_{b|y}.$$
Then
$$p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, \Pi_{ab|xy}\big) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big)$$
is a probability law on $\{\pm1\} \times \{\pm1\}$, and
$$\sum_{b=\pm1} p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x}\big), \qquad \sum_{a=\pm1} p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(R)}_{b|y}\big).$$
Hence the local marginal on one wing is independent of the remote setting.

Proof sketch. Because the left/right projectors commute, $\{\Pi_{ab|xy}\}_{a,b}$ is a commuting projective family on the fixed setting slice. The Born trace therefore gives the joint law. Positivity is immediate, normalization follows from the completeness relations, and summing over one wing removes the remote projector.

Theorem 3.8 (CHSH/Tsirelson on the fixed-cutoff Bell surface). Define
$$A_x := P^{(L)}_{+|x} - P^{(L)}_{-|x}, \qquad B_y := P^{(R)}_{+|y} - P^{(R)}_{-|y}.$$
Then
$$A_x^2 = \mathbf{1}, \qquad B_y^2 = \mathbf{1}, \qquad [A_x, B_y] = 0,$$
and the Bell correlators satisfy
$$E(x, y) := \sum_{a,b=\pm1} ab\, p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big).$$

For the CHSH operator

$$\mathcal{B} := A_0(B_0 + B_1) + A_1(B_0 - B_1)$$

one has

$$\mathcal{B}^2 = 4\,\mathbf{1} - [A_0, A_1][B_0, B_1], \qquad \|\mathcal{B}\| \leq 2\sqrt{2}.$$

Therefore

$$\big|E(0,0) + E(0,1) + E(1,0) - E(1,1)\big| \leq 2\sqrt{2}.$$

Proof sketch. Expand $A_x B_y$ in the projective basis to get the correlator formula. Expand $\mathcal{B}^2$ and use $A_x^2 = B_y^2 = \mathbf{1}$ together with $[A_x, B_y] = 0$. Since $\|[A_0, A_1]\| \leq 2$ and $\|[B_0, B_1]\| \leq 2$, the operator norm of $\mathcal{B}^2$ is at most 8.

Corollary 3.9 (Two-qubit saturation branch). Assume additionally that there exist commuting unital $*$-subalgebras

$$\mathcal{Q}_L \subset \mathcal{A}_L, \qquad \mathcal{Q}_R \subset \mathcal{A}_R,$$

with

$$\mathcal{Q}_L \cong M_2(\mathbb{C}), \qquad \mathcal{Q}_R \cong M_2(\mathbb{C}), \qquad C^*(\mathcal{Q}_L, \mathcal{Q}_R) \cong M_2(\mathbb{C}) \otimes M_2(\mathbb{C}).$$

Assume that, under this identification, the restriction of ρ_{LR} to $C^*(\mathcal{Q}_L, \mathcal{Q}_R)$ is

$$|\Phi^+\rangle\langle\Phi^+|, \qquad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

and that the fixed-cutoff observables restrict there to

$$A_0 = \sigma_z \otimes \mathbf{1}, \qquad A_1 = \sigma_x \otimes \mathbf{1},$$

$$B_0 = \mathbf{1} \otimes \frac{\sigma_z + \sigma_x}{\sqrt{2}}, \qquad B_1 = \mathbf{1} \otimes \frac{\sigma_z - \sigma_x}{\sqrt{2}}.$$

Then

$$E(0,0) = E(0,1) = E(1,0) = \frac{1}{\sqrt{2}}, \qquad E(1,1) = -\frac{1}{\sqrt{2}},$$

and hence

$$E(0,0) + E(0,1) + E(1,0) - E(1,1) = 2\sqrt{2}.$$

Proof sketch. This is the standard two-qubit computation on the explicitly assumed $M_2(\mathbb{C}) \otimes M_2(\mathbb{C})$ subalgebra. It is an exact branch-level statement on that added two-qubit input; it does not derive tensor-factor structure or Bell-pair preparation from the bare OPH repair dynamics.